

EEE 263 – ELECTRONIC CIRCUITS

Metal Oxide Semiconductor Field Effect Transistor

Khairul Islam

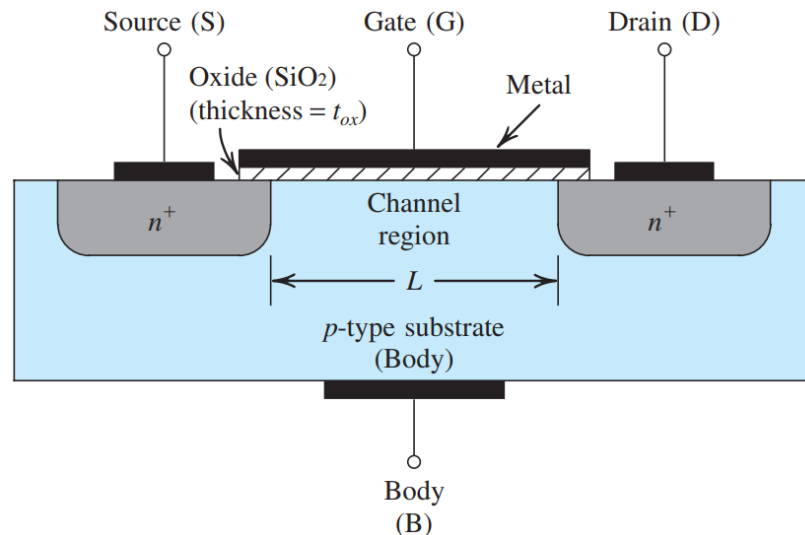
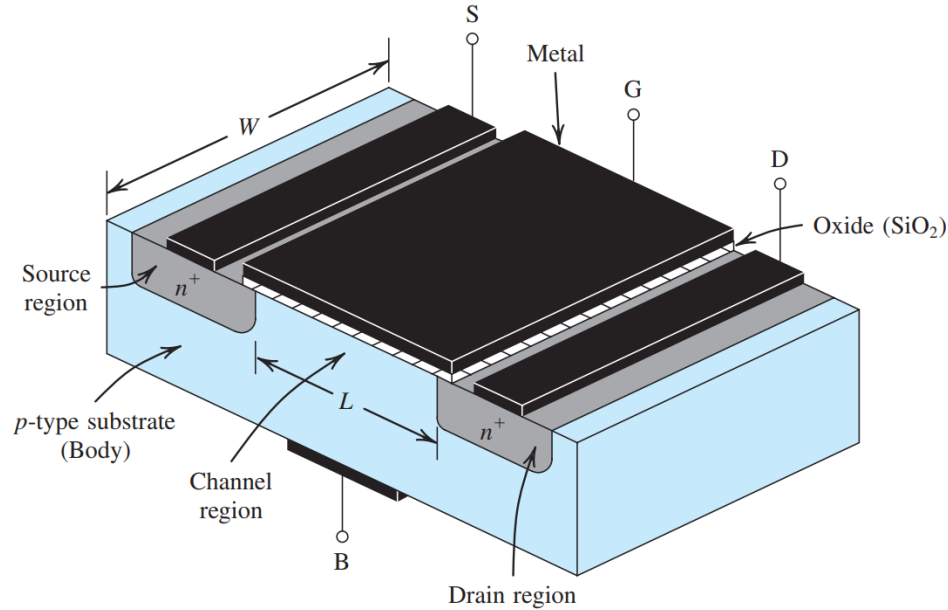
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Device Structure



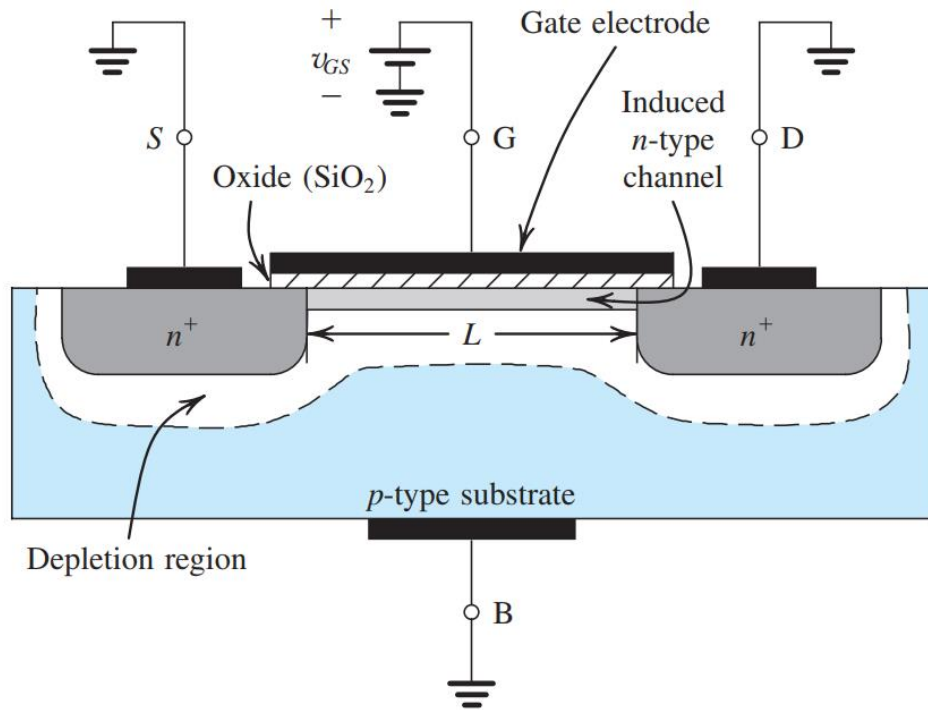
- Two types: **n-MOS, p-MOS**
- Four terminals:
Gate, Source, Drain, Body
- Insulator SiO₂ with metal contact form **gate electrode**
- Source → n^+ (p^+)
- Drain → n^+ (p^+)
- Substrate → p (n)
- When body is connected to source,
3 terminals: G, S, D

Operating Principle

With zero gate voltage:

- Two back to back diode
- No current flow even V_{DS} is applied due to high resistance ($\cong 10^{12}\Omega$)

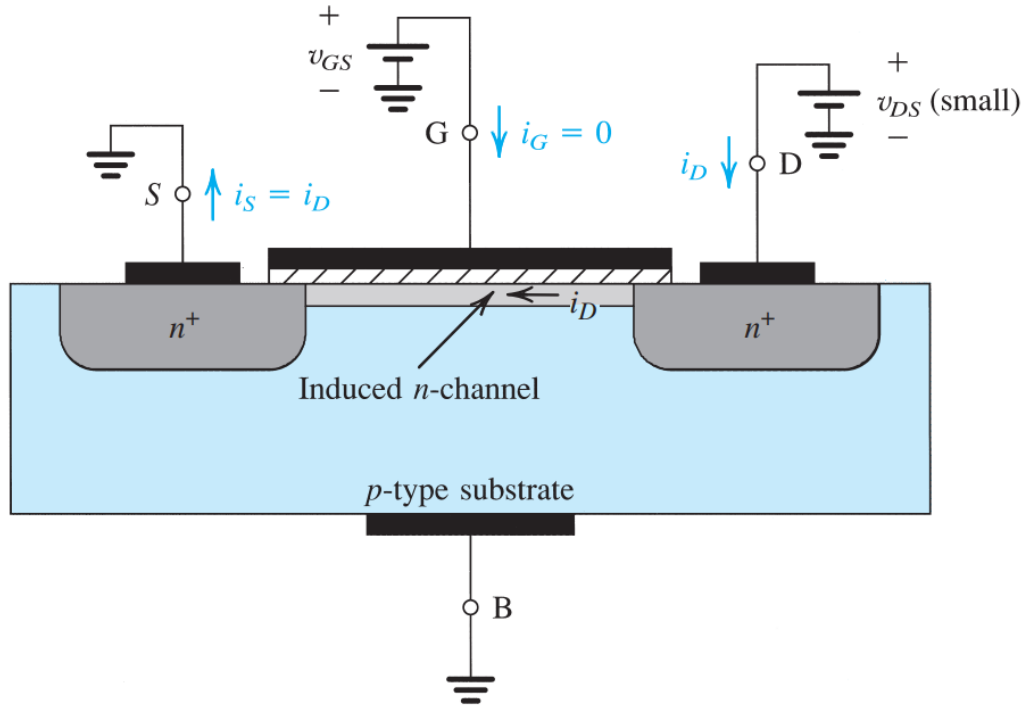
Creating a channel:



- Holes pushed downward
- Depletion region creates
- Electrons are attracted
- n region created forms a channel $\rightarrow$ n-channel which is called **Inversion Layer**
- Threshold voltage, V_t at which conducting channel forms
- Oxide layer $\rightarrow$ Capacitor dielectric & field creates
- Voltage along channel 0 & across oxide V_{GS}

Operating Principle

Applying a small v_{DS} :



Effective Voltage or Overdrive Voltage: $v_{GS} - V_t \equiv v_{OV}$

Charge in channel: $|Q| = C_{ox}(WL)v_{OV}$

where C_{ox} , called the **oxide capacitance**, is the capacitance of the parallel-plate capacitor per unit gate area, $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

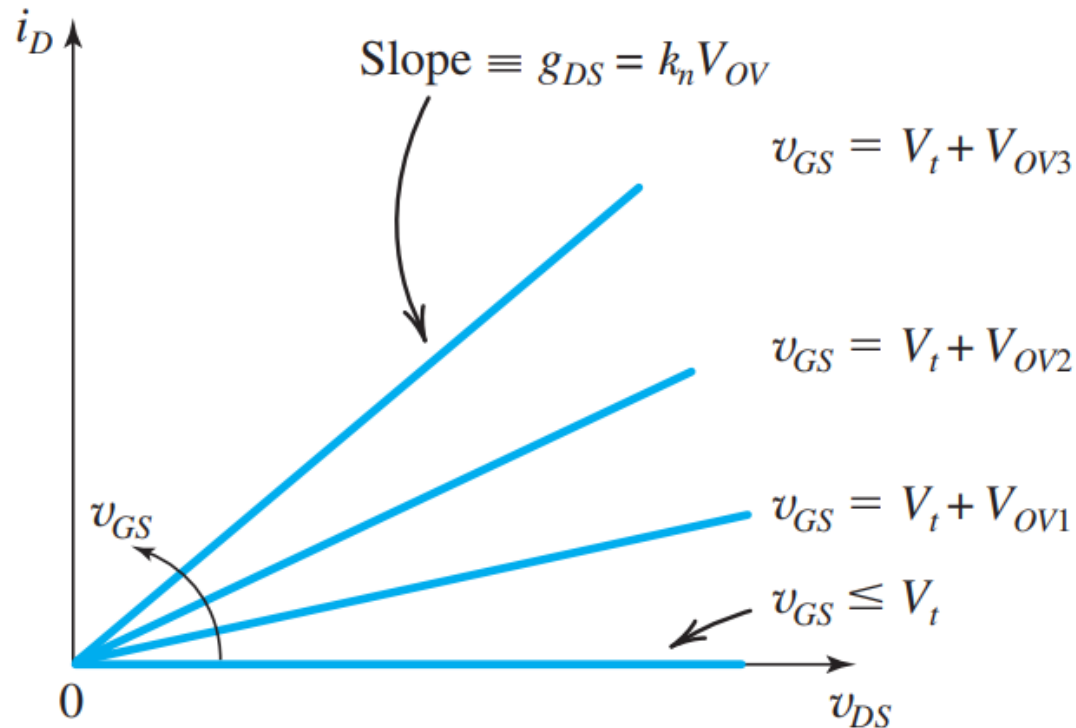
where ϵ_{ox} is the permittivity of the silicon dioxide,

$$\epsilon_{ox} = 3.9\epsilon_0 = 3.9 \times 8.854 \times 10^{-12} = 3.45 \times 10^{-11} \text{ F/m}$$

Total capacitance between gate and channel: $C = C_{ox}WL$

Operating Principle

Applying a small v_{DS} :



$$i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t) \right] v_{DS}$$

Total conductance of the channel:

$$g_{DS} = (\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t)$$

The process transconductance parameter: $k'_n = \mu_n C_{ox}$

The transistor aspect ratio: (W/L)

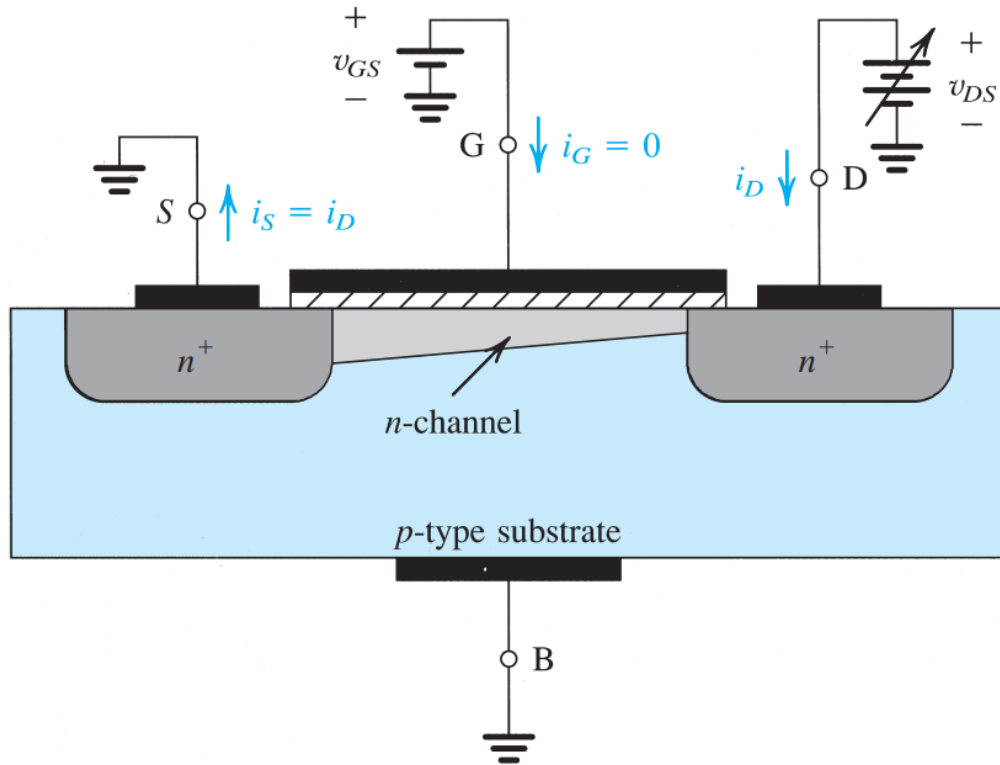
The MOSFET transconductance parameter:

$$k_n = k'_n (W/L) = (\mu_n C_{ox}) (W/L)$$

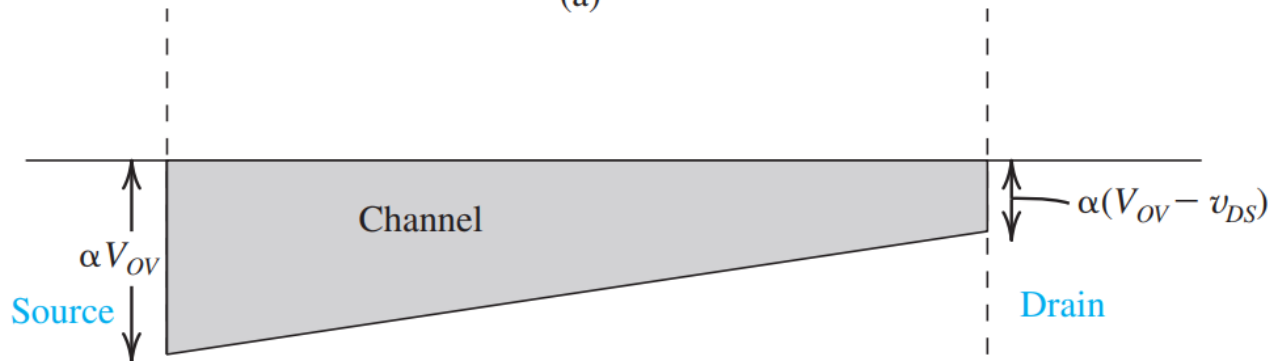
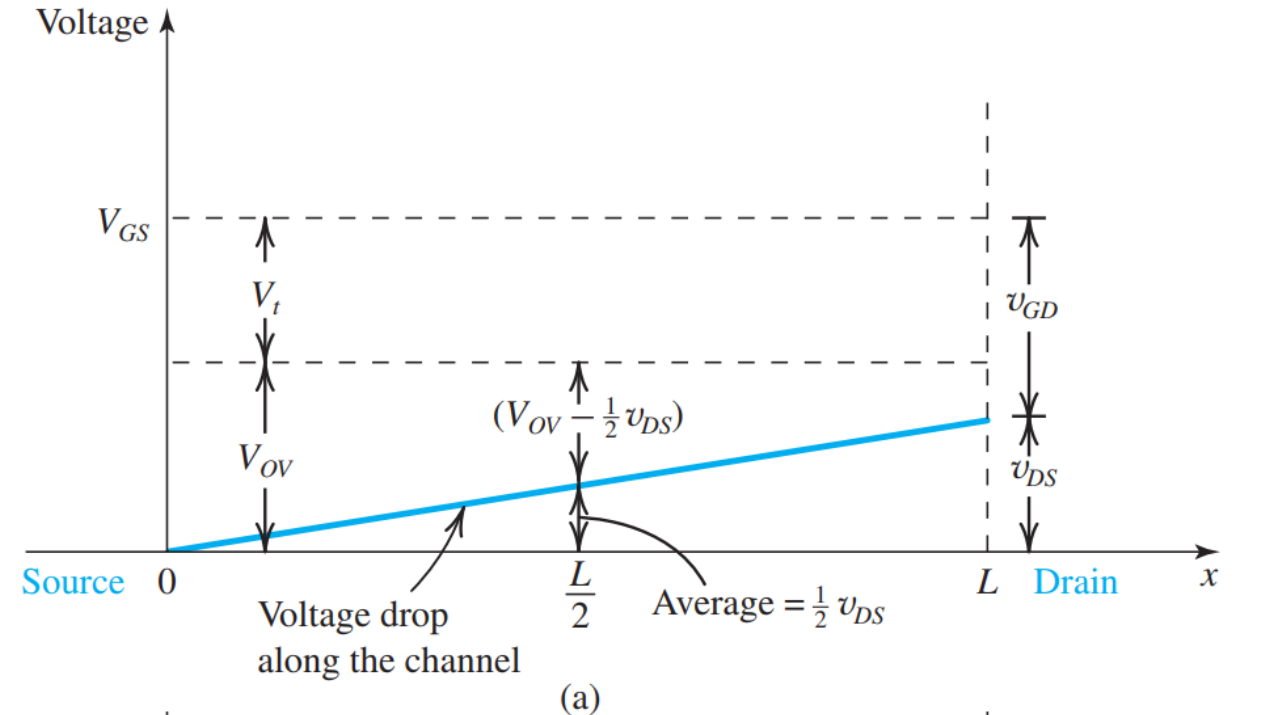
$$r_{DS} = \frac{1}{g_{DS}} = \frac{1}{(\mu_n C_{ox}) (W/L) (v_{GS} - V_t)}$$

Operating Principle

Operation as v_{DS} is increased ($v_{DS} \leq V_{OV}$):



$$i_D = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$



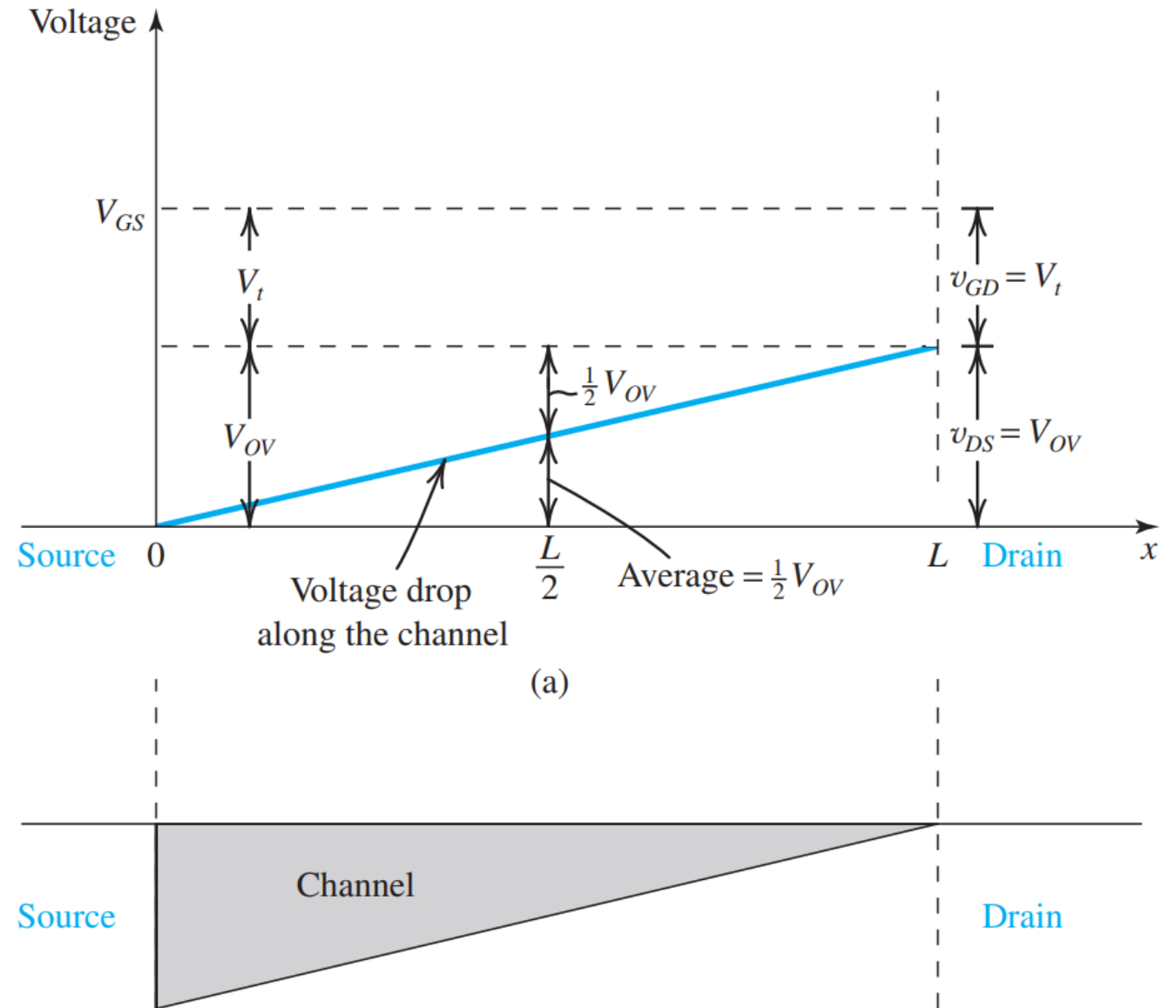
Operating Principle

Operation as v_{DS} is equal to V_{OV} :

$$V_{DSsat} = V_{OV} = V_{GS} - V_t$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{OV}^2$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_t)^2$$



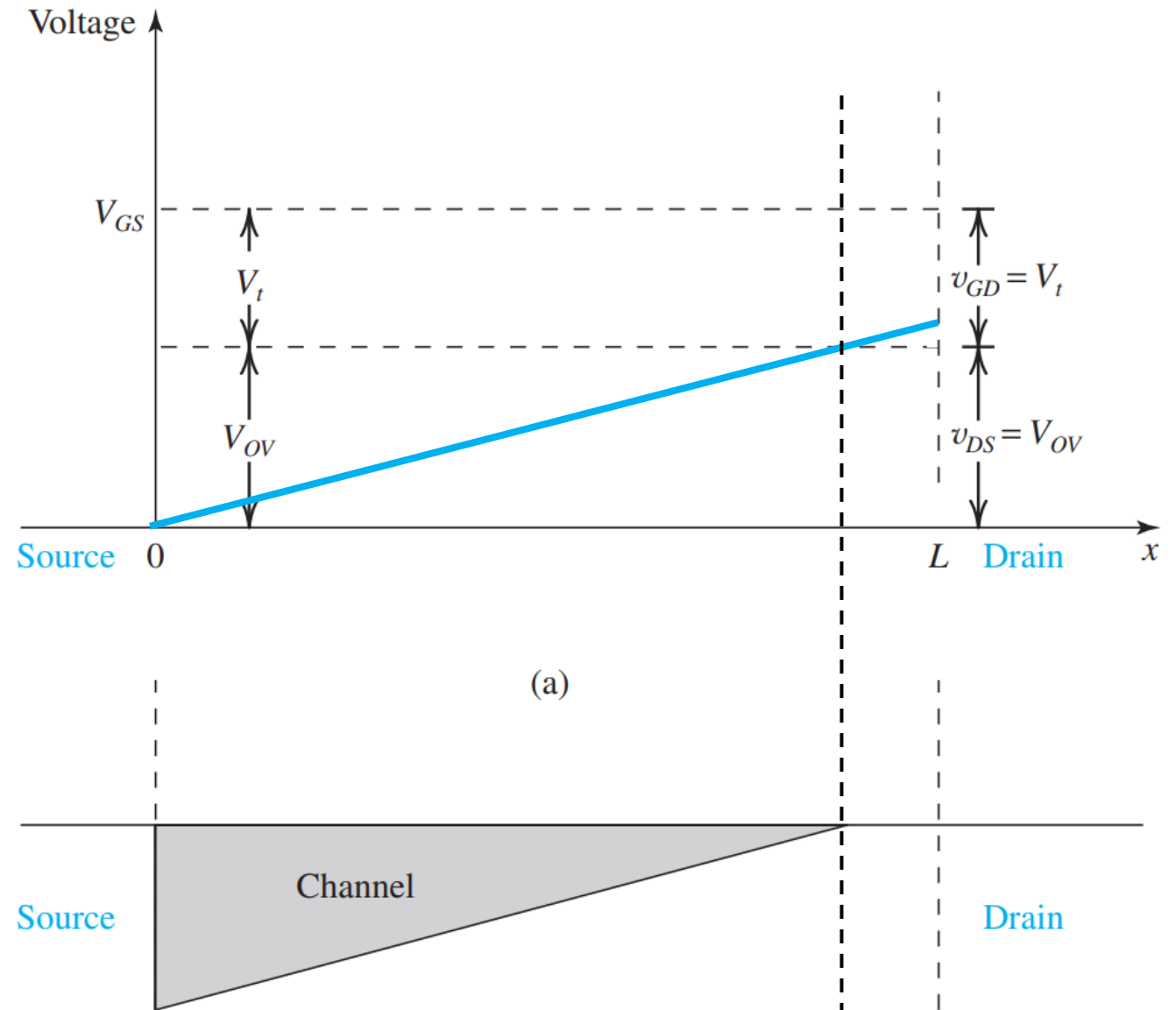
Operating Principle

Operation as v_{DS} is increased beyond V_{OV} :

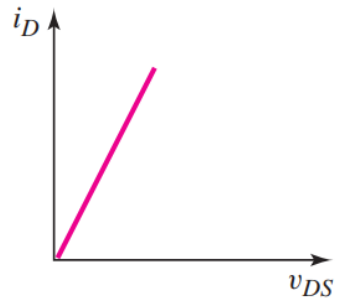
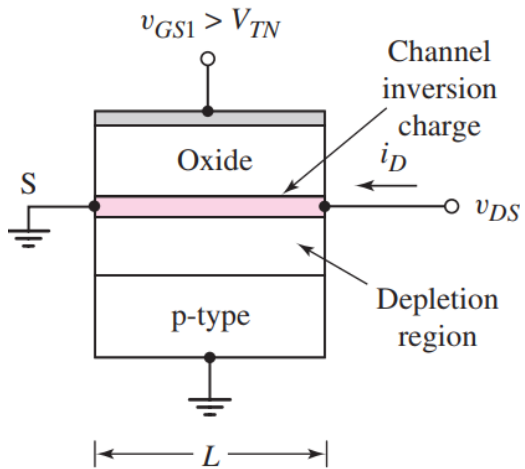
$$V_{DSsat} = V_{OV} = V_{GS} - V_t$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{OV}^2$$

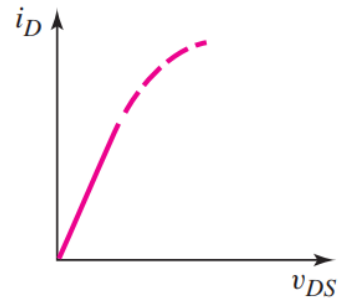
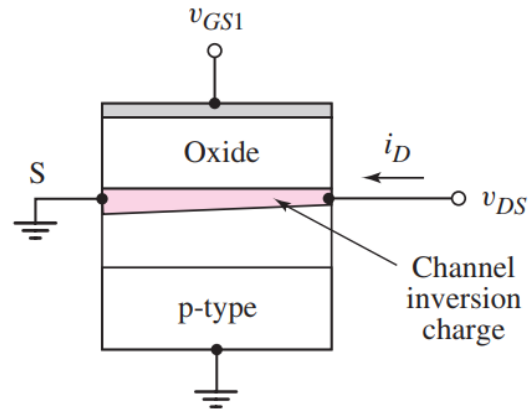
$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_t)^2$$



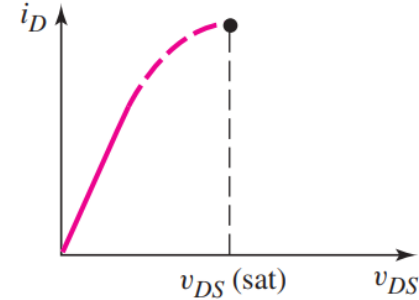
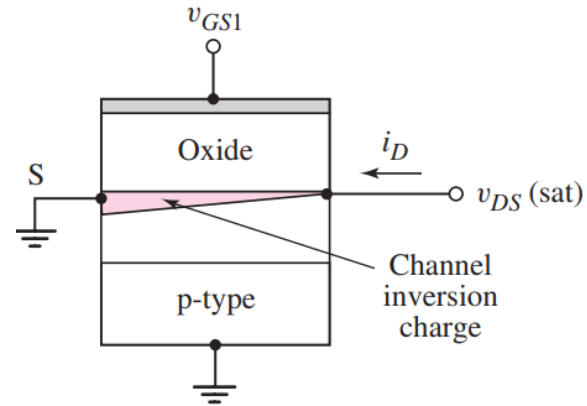
Operating Principle



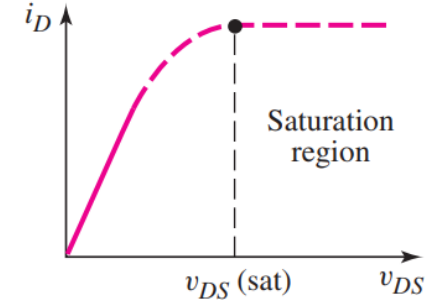
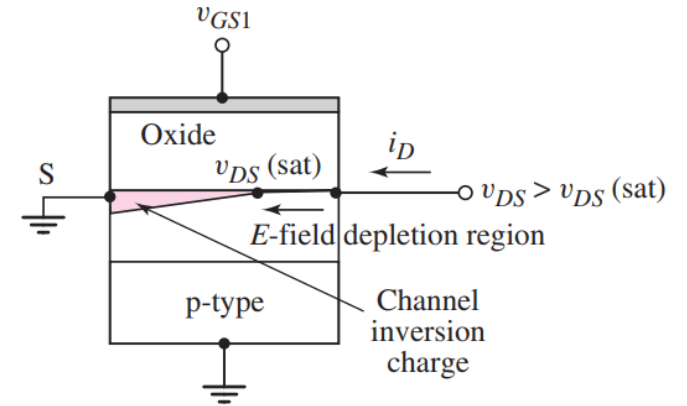
(a)



(b)

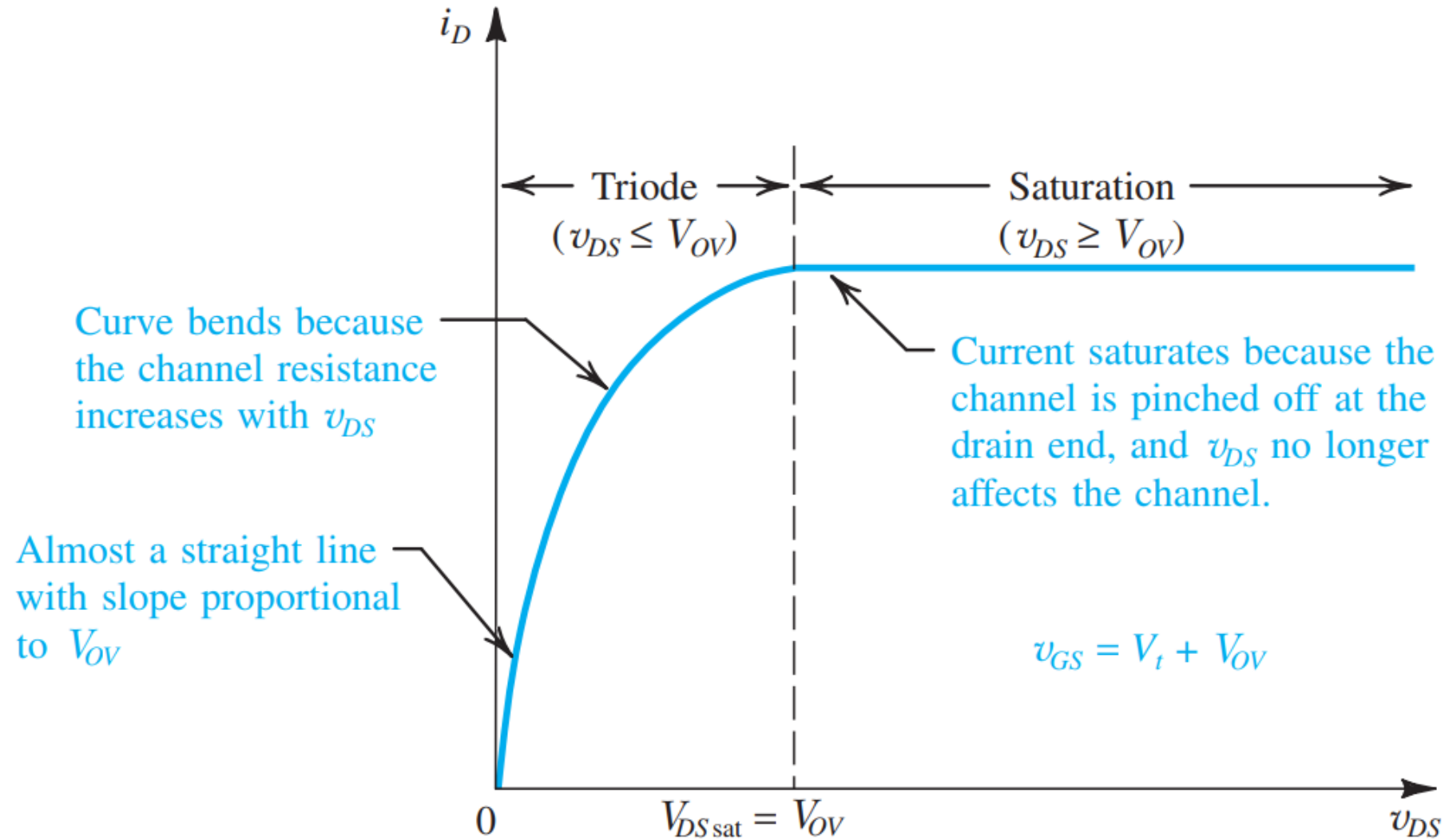


(c)



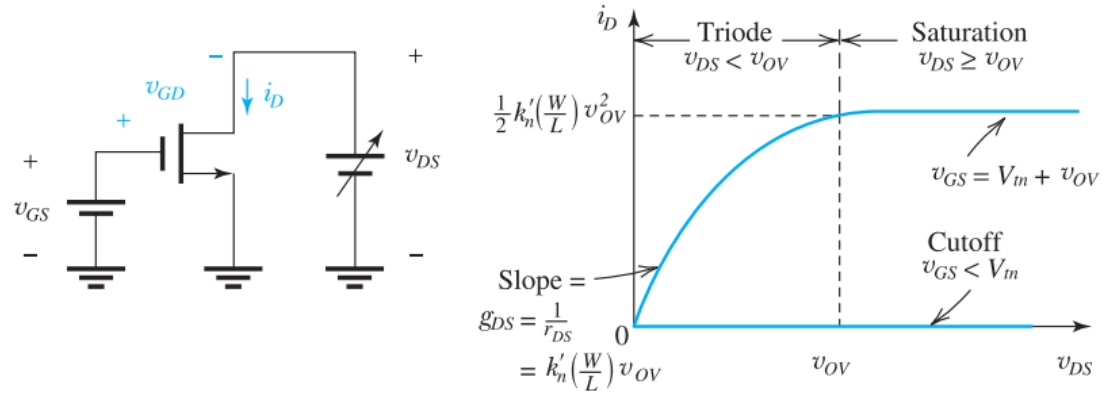
(d)

Operating Principle



Operating Principle

Regions of Operation of the Enhancement NMOS Transistor



- $v_{GS} < V_{tm}$: no channel; transistor in cutoff; $i_D = 0$
- $v_{GS} = V_{tm} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{GD} > V_{tm}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_{tm})v_{DS} - \frac{1}{2}v_{DS}^2 \right]$$

or equivalently,

$$i_D = k'_n \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2}v_{DS} \right) v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tm}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

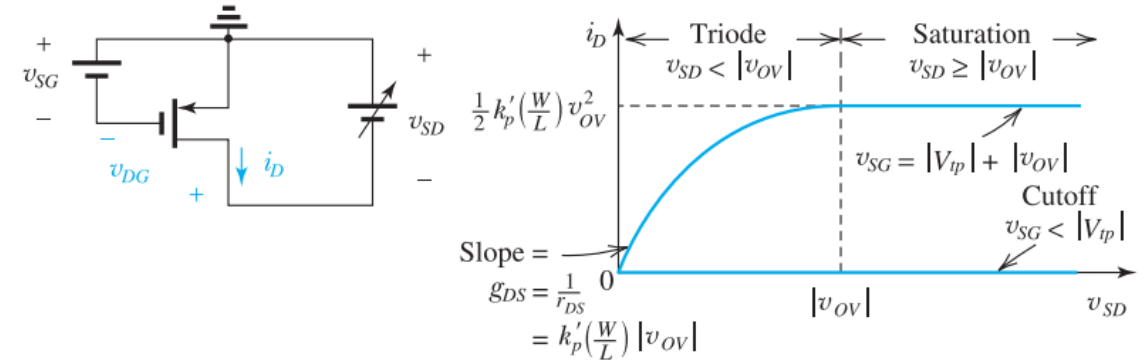
Then

$$i_D = \frac{1}{2}k'_n \left(\frac{W}{L} \right) (v_{GS} - V_{tm})^2$$

or equivalently,

$$i_D = \frac{1}{2}k'_n \left(\frac{W}{L} \right) v_{OV}^2$$

Regions of Operation of the Enhancement PMOS Transistor



- $v_{SG} < |V_{tp}|$: no channel; transistor in cutoff; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently

$$v_{SD} < |v_{OV}|$$

Then

$$i_D = k'_p \left(\frac{W}{L} \right) \left[(v_{SG} - |V_{tp}|)v_{SD} - \frac{1}{2}v_{SD}^2 \right]$$

or equivalently

$$i_D = k'_p \left(\frac{W}{L} \right) \left(|v_{OV}| - \frac{1}{2}v_{SD} \right) v_{SD}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

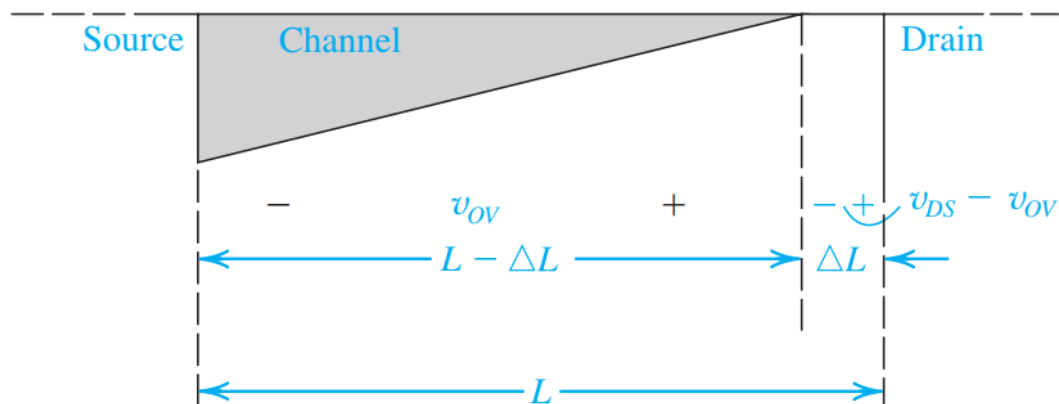
Then

$$i_D = \frac{1}{2}k'_p \left(\frac{W}{L} \right) (v_{SG} - |V_{tp}|)^2$$

or equivalently

$$i_D = \frac{1}{2}k'_p \left(\frac{W}{L} \right) v_{OV}^2$$

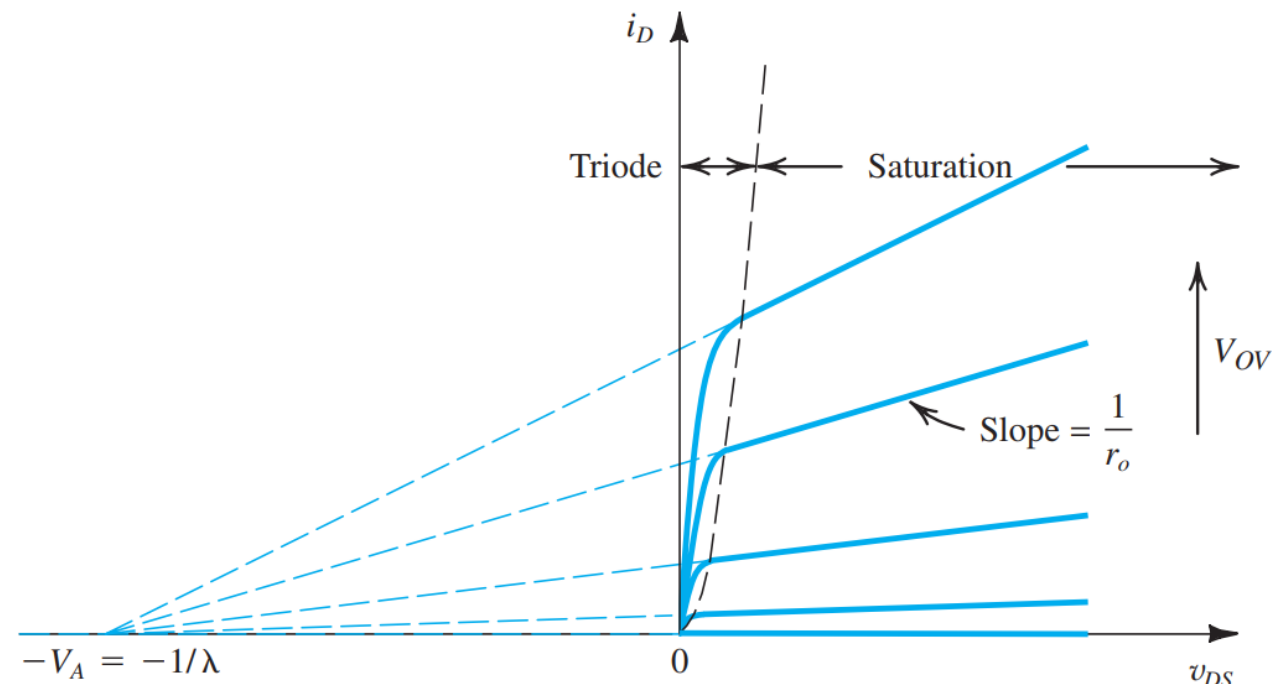
Channel-Length Modulation



$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2 (1 + \lambda v_{DS})$$

$$V_A = \frac{1}{\lambda}$$

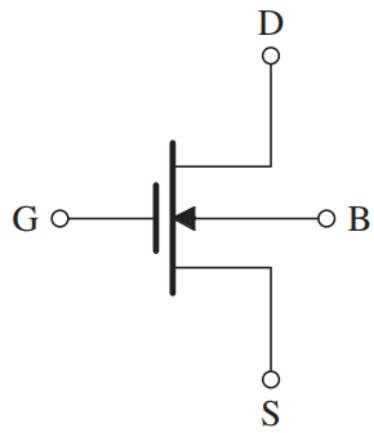
$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}} \right]_{v_{GS} \text{ constant}}^{-1} = \left[\lambda \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_{tn})^2 \right]^{-1} = \frac{1}{\lambda I'_D} = \frac{V_A}{I'_D}$$



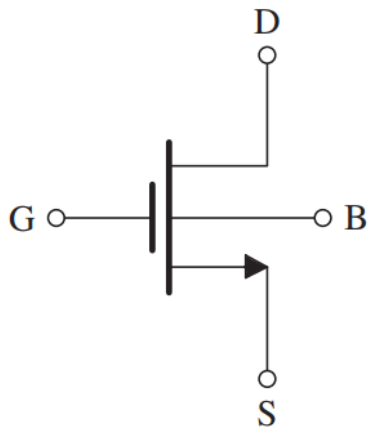
where I'_D is the drain current *without* channel-length modulation taken into account;

$$I'_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{tn})^2$$

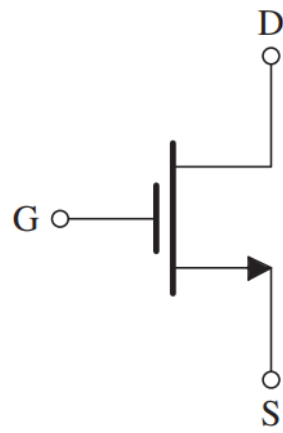
Circuit Symbols and Equivalent Circuit



(a)

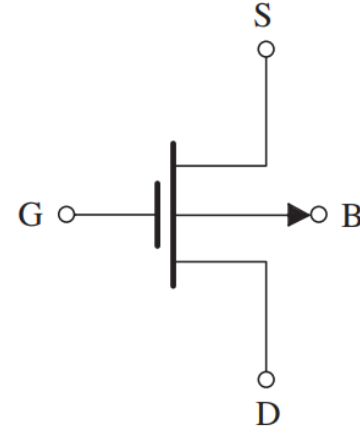


(b)

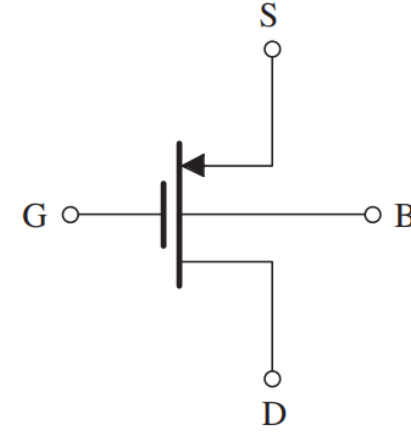


(c)

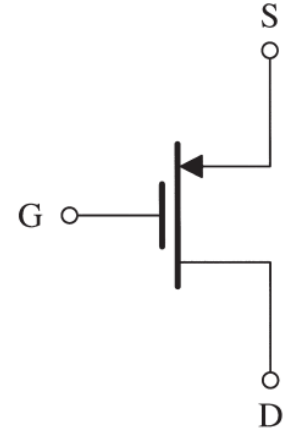
Circuit symbol for the *n*-channel enhancement-type MOSFET.



(a)

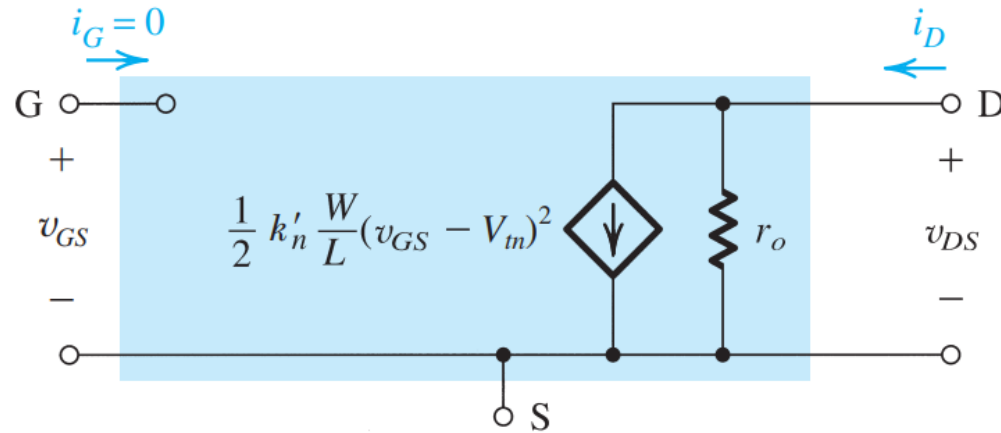


(b)



(c)

Circuit symbol for the *p*-channel enhancement-type MOSFET.



Large-signal, equivalent-circuit model of the *n*-channel MOSFET in saturation

Examples

Consider a process technology for which $L_{\min} = 0.4 \mu\text{m}$, $t_{ox} = 8 \text{ nm}$, $\mu_n = 450 \text{ cm}^2/\text{V} \cdot \text{s}$, and $V_t = 0.7 \text{ V}$.

- (a) Find C_{ox} and k'_n .
- (b) For a MOSFET with $W/L = 8 \mu\text{m}/0.8 \mu\text{m}$, calculate the values of V_{OV} , V_{GS} , and $V_{DS\min}$ needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu\text{A}$.
- (c) For the device in (b), find the values of V_{OV} and V_{GS} required to cause the device to operate as a $1000\text{-}\Omega$ resistor for very small v_{DS} .

Solution

(a)

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{8 \times 10^{-9}} = 4.32 \times 10^{-3} \text{ F/m}^2$$
$$= 4.32 \text{ fF}/\mu\text{m}^2$$

$$k'_n = \mu_n C_{ox} = 450 (\text{cm}^2/\text{V} \cdot \text{s}) \times 4.32 (\text{fF}/\mu\text{m}^2)$$
$$= 450 \times 10^8 (\mu\text{m}^2/\text{V} \cdot \text{s}) \times 4.32 \times 10^{-15} (\text{F}/\mu\text{m}^2)$$
$$= 194 \times 10^{-6} (\text{F}/\text{V} \cdot \text{s})$$
$$= 194 \mu\text{A}/\text{V}^2$$

Examples

(b) For operation in the saturation region,

$$i_D = \frac{1}{2} k'_n \frac{W}{L} V_{OV}^2$$

Thus,

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} V_{OV}^2$$

which results in

$$V_{OV} = 0.32 \text{ V}$$

Thus,

$$V_{GS} = V_t + V_{OV} = 1.02 \text{ V}$$

and

$$V_{DSmin} = V_{OV} = 0.32 \text{ V}$$

(c) For the MOSFET in the triode region with v_{DS} very small,

$$r_{DS} = \frac{1}{k'_n \frac{W}{L} V_{OV}}$$

Thus

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 \times V_{OV}}$$

which yields

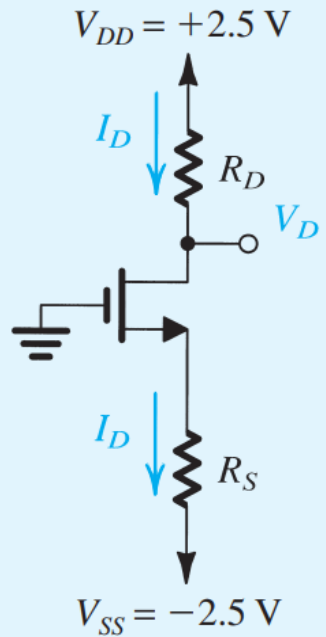
$$V_{OV} = 0.52 \text{ V}$$

Thus,

$$V_{GS} = 1.22 \text{ V}$$

MOSFET Circuits at DC

Design the circuit of Fig. 5.21: that is, determine the values of R_D and R_S so that the transistor operates at $I_D = 0.4 \text{ mA}$ and $V_D = +0.5 \text{ V}$. The NMOS transistor has $V_t = 0.7 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $L = 1 \mu\text{m}$, and $W = 32 \mu\text{m}$. Neglect the channel-length modulation effect (i.e., assume that $\lambda = 0$).



$$R_D = \frac{V_{DD} - V_D}{I_D} \\ = \frac{2.5 - 0.5}{0.4} = 5 \text{ k}\Omega$$

which results in

Thus,

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{OV}^2 \\ 400 = \frac{1}{2} \times 100 \times \frac{32}{1} V_{OV}^2$$

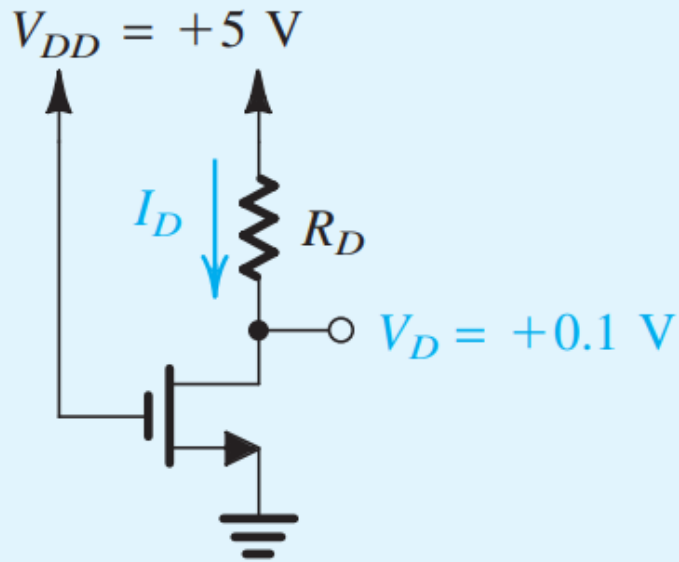
$$V_{OV} = 0.5 \text{ V}$$

$$V_{GS} = V_t + V_{OV} = 0.7 + 0.5 = 1.2 \text{ V}$$

$$R_S = \frac{V_S - V_{SS}}{I_D} \\ = \frac{-1.2 - (-2.5)}{0.4} = 3.25 \text{ k}\Omega$$

MOSFET Circuits at DC

Design the circuit in Fig. 5.23 to establish a drain voltage of 0.1 V. What is the effective resistance between drain and source at this operating point? Let $V_{in} = 1$ V and $k'_n(W/L) = 1$ mA/V².



$$I_D = k'_n \frac{W}{L} \left[(V_{GS} - V_{in}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$
$$I_D = 1 \times \left[(5 - 1) \times 0.1 - \frac{1}{2} \times 0.01 \right]$$
$$= 0.395 \text{ mA}$$

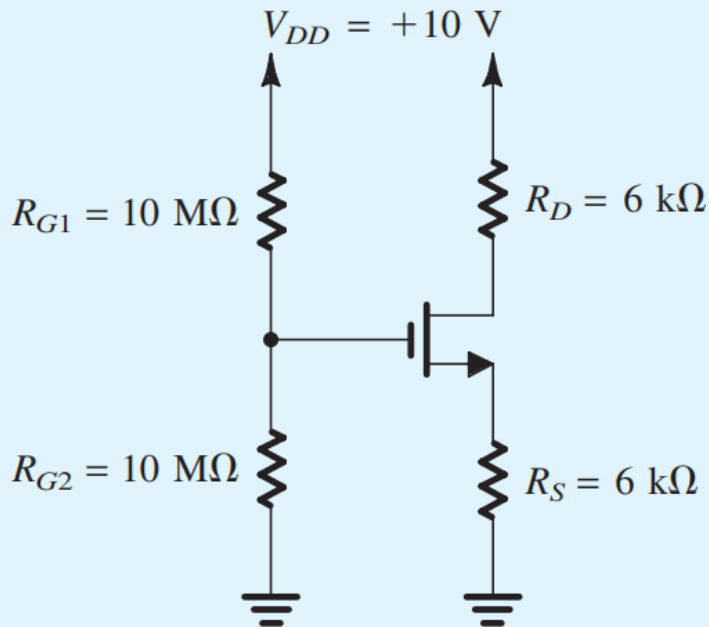
$$R_D = \frac{V_{DD} - V_D}{I_D}$$
$$= \frac{5 - 0.1}{0.395} = 12.4 \text{ k}\Omega$$

$$r_{DS} = \frac{V_{DS}}{I_D}$$
$$= \frac{0.1}{0.395} = 253 \text{ }\Omega$$

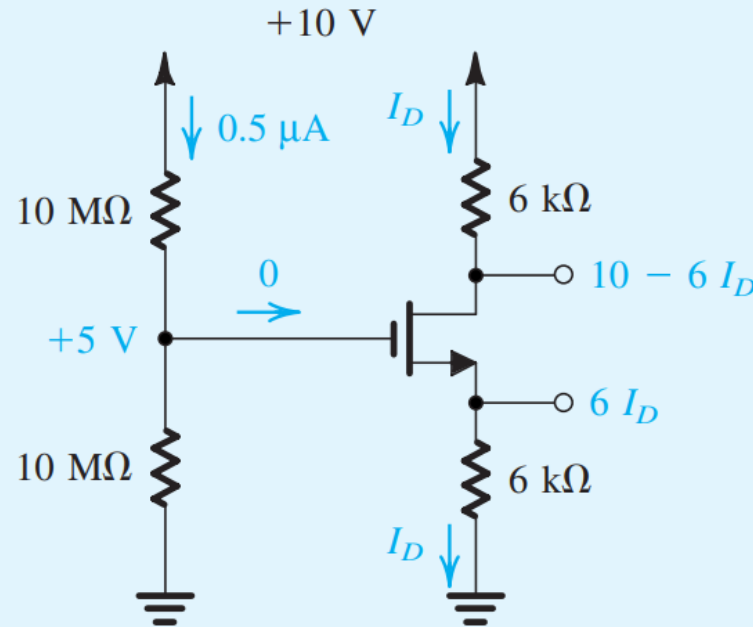
$$r_{DS} = \frac{1}{k'_n V_{OV}}$$
$$r_{DS} = \frac{1}{1 \times (5 - 1)} = 0.25 \text{ k}\Omega = 250 \text{ }\Omega$$

MOSFET Circuits at DC

Analyze the circuit shown in Fig. 5.24(a) to determine the voltages at all nodes and the currents through all branches. Let $V_m = 1$ V and $k'_n(W/L) = 1$ mA/V². Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).



(a)



(b)

which results in the following quadratic equation in I_D :

$$18I_D^2 - 25I_D + 8 = 0$$

$$V_G = V_{DD} \frac{R_{G2}}{R_{G2} + R_{G1}} = 10 \times \frac{10}{10 + 10} = +5\text{ V}$$

$$V_{GS} = 5 - 6I_D$$

$$I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_m)^2$$

$$= \frac{1}{2} \times 1 \times (5 - 6I_D - 1)^2$$

$$I_D = 0.5\text{ mA}$$

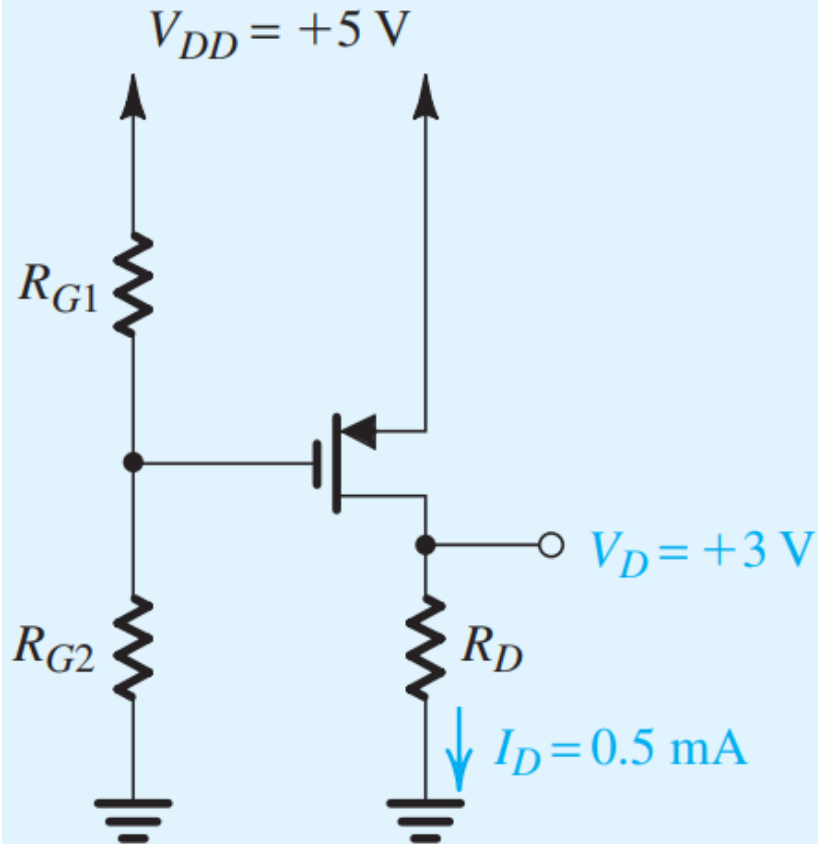
$$V_S = 0.5 \times 6 = +3\text{ V}$$

$$V_{GS} = 5 - 3 = 2\text{ V}$$

$$V_D = 10 - 6 \times 0.5 = +7\text{ V}$$

MOSFET Circuits at DC

Design the circuit of Fig. 5.25 so that the transistor operates in saturation with $I_D = 0.5 \text{ mA}$ and $V_D = +3 \text{ V}$. Let the PMOS transistor have $V_{tp} = -1 \text{ V}$ and $k'_p(W/L) = 1 \text{ mA/V}^2$. Assume $\lambda = 0$. What is the largest value that R_D can have while maintaining saturation-region operation?



$$|V_{ov}| = 1 \text{ V}$$

$$V_{SG} = |V_{tp}| + |V_{ov}| = 1 + 1 = 2 \text{ V}$$

$$R_D = \frac{V_D}{I_D} = \frac{3}{0.5} = 6 \text{ k}\Omega$$

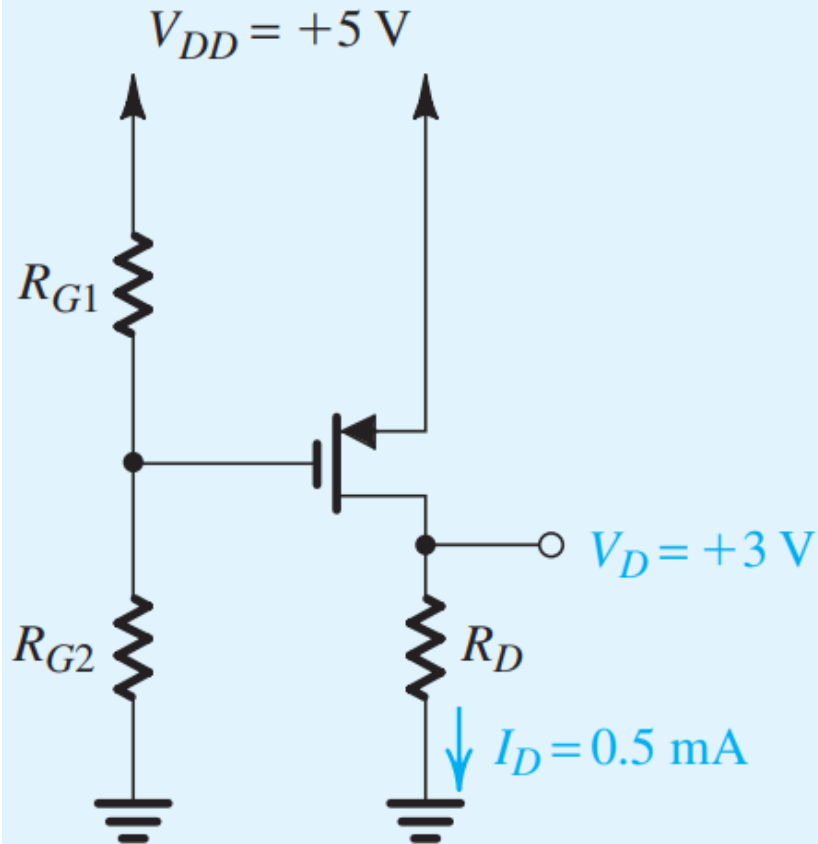
$$V_{D_{\max}} = 3 + 1 = 4 \text{ V}$$

This value of drain voltage is obtained with R_D given by

$$R_D = \frac{4}{0.5} = 8 \text{ k}\Omega$$

MOSFET Circuits at DC

Design the circuit of Fig. 5.25 so that the transistor operates in saturation with $I_D = 0.5 \text{ mA}$ and $V_D = +3 \text{ V}$. Let the PMOS transistor have $V_{tp} = -1 \text{ V}$ and $k'_p(W/L) = 1 \text{ mA/V}^2$. Assume $\lambda = 0$. What is the largest value that R_D can have while maintaining saturation-region operation?



$$|V_{ov}| = 1 \text{ V}$$

$$V_{SG} = |V_{tp}| + |V_{ov}| = 1 + 1 = 2 \text{ V}$$

$$R_D = \frac{V_D}{I_D} = \frac{3}{0.5} = 6 \text{ k}\Omega$$

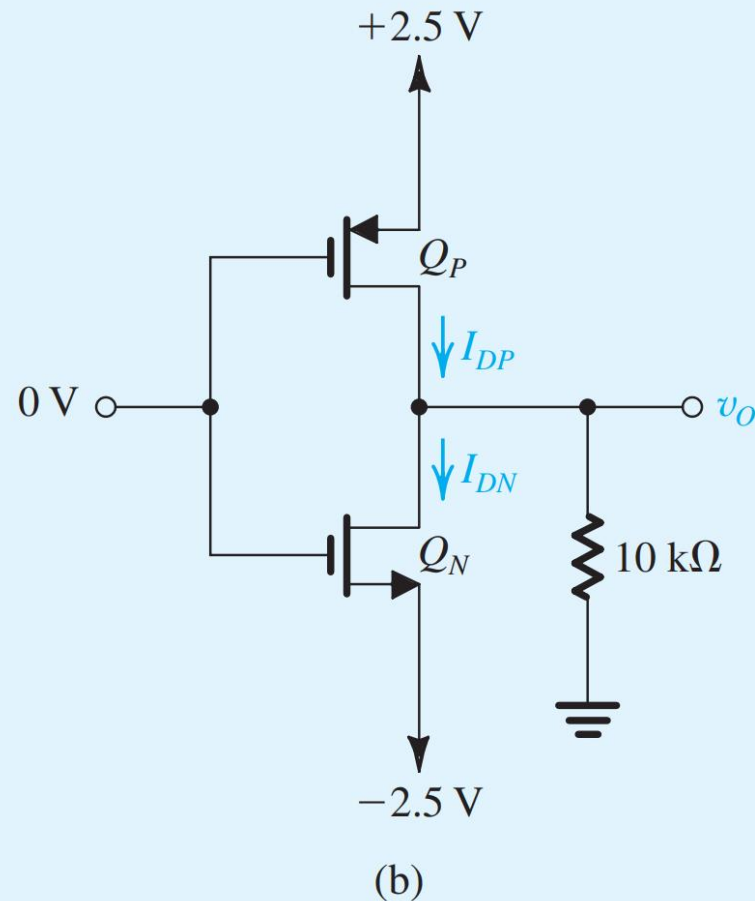
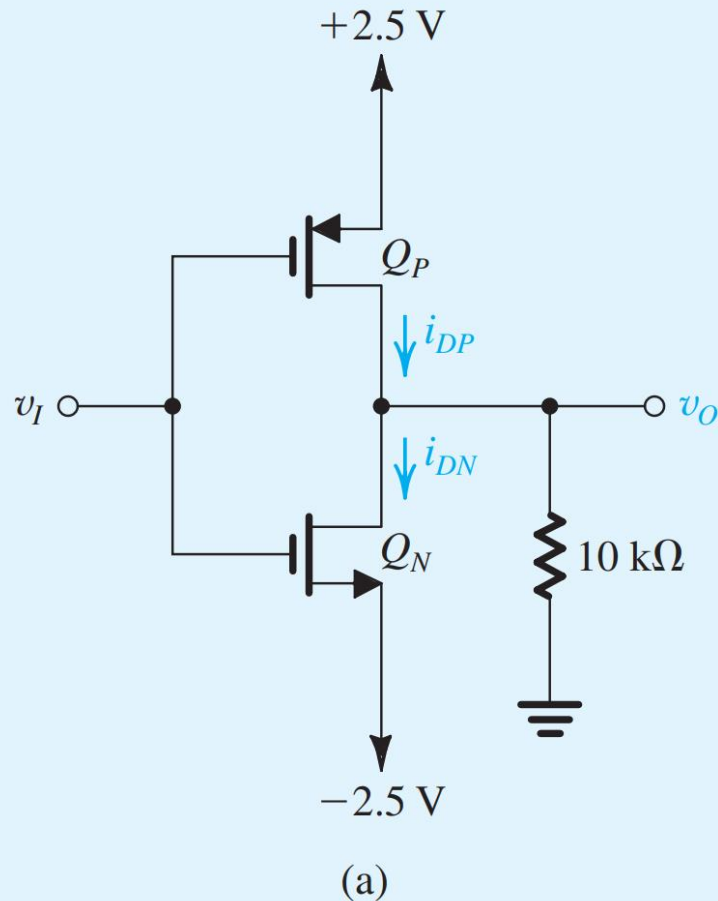
$$V_{D_{\max}} = 3 + 1 = 4 \text{ V}$$

This value of drain voltage is obtained with R_D given by

$$R_D = \frac{4}{0.5} = 8 \text{ k}\Omega$$

MOSFET Circuits at DC

The NMOS and PMOS transistors in the circuit of Fig. 5.26(a) are matched, with $k'_n(W_n/L_n) = k'_p(W_p/L_p) = 1 \text{ mA/V}^2$ and $V_{tn} = -V_{tp} = 1 \text{ V}$. Assuming $\lambda = 0$ for both devices, find the drain currents i_{DN} and i_{DP} , as well as the voltage v_O , for $v_I = 0 \text{ V}$, $+2.5 \text{ V}$, and -2.5 V .



MOSFET Circuits at DC

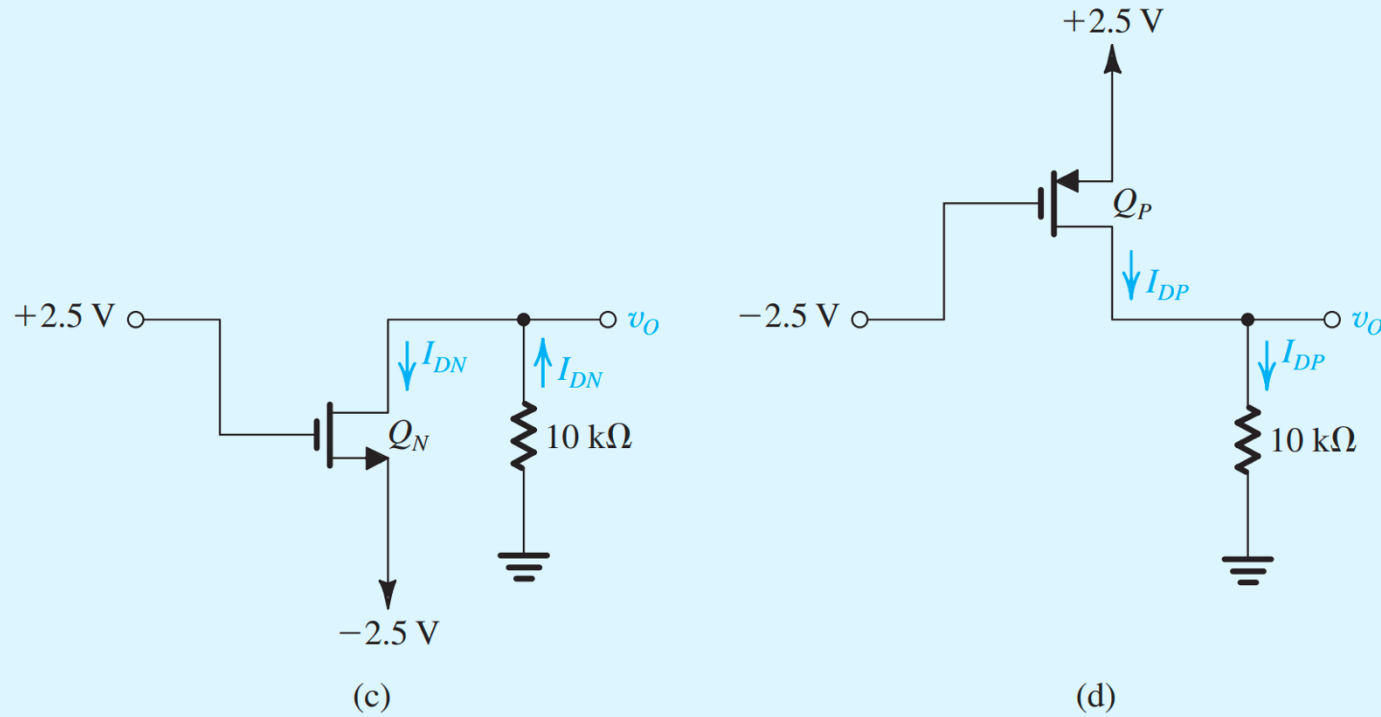


Figure 5.26 continued

Solution

Figure 5.26(b) shows the circuit for the case $v_I = 0$ V. We note that since Q_N and Q_P are perfectly matched and are operating at equal values of $|V_{GS}| = 2.5$ V, the circuit is symmetrical, which dictates that $v_O = 0$ V. Thus both Q_N and Q_P are operating with $|V_{DG}| = 0$ and, hence, in saturation. The drain currents can now be found from

$$I_{DP} = I_{DN} = \frac{1}{2} \times 1 \times (2.5 - 1)^2 = 1.125 \text{ mA}$$

MOSFET Circuits at DC

Next, we consider the circuit with $v_I = +2.5$ V. Transistor Q_P will have a V_{SG} of zero and thus will be cut off, reducing the circuit to that shown in Fig. 5.26(c). We note that v_O will be negative, and thus v_{GD} will be greater than V_m , causing Q_N to operate in the triode region. For simplicity we shall assume that v_{DS} is small and thus use

$$\begin{aligned} I_{DN} &\simeq k'_n (W_n/L_n) (V_{GS} - V_m) V_{DS} \\ &= 1[2.5 - (-2.5) - 1][v_O - (-2.5)] \end{aligned}$$

From the circuit diagram shown in Fig. 5.26(c), we can also write

$$I_{DN}(\text{mA}) = \frac{0 - v_O}{10(\text{k}\Omega)}$$

These two equations can be solved simultaneously to yield

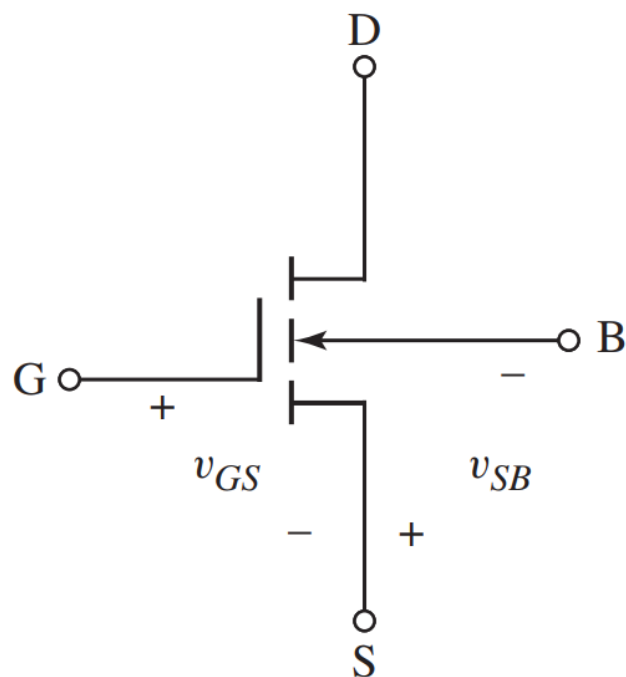
$$I_{DN} = 0.244 \text{ mA} \quad v_O = -2.44 \text{ V}$$

Note that $V_{DS} = -2.44 - (-2.5) = 0.06$ V, which is small as assumed.

Finally, the situation for the case $v_I = -2.5$ V [Fig. 5.26(d)] will be the exact complement of the case $v_I = +2.5$ V: Transistor Q_N will be off. Thus $I_{DN} = 0$, Q_P will be operating in the triode region with $I_{DP} = 0.244$ mA and $v_O = +2.44$ V.



The Body Effect



An n-channel enhancement mode MOSFET with a substrate voltage

The effect of V_{SB} on the channel can be most conveniently represented as a change in the threshold voltage V_t . Specifically, it has been shown that increasing the reverse substrate bias voltage V_{SB} results in an increase in V_t according to the relationship

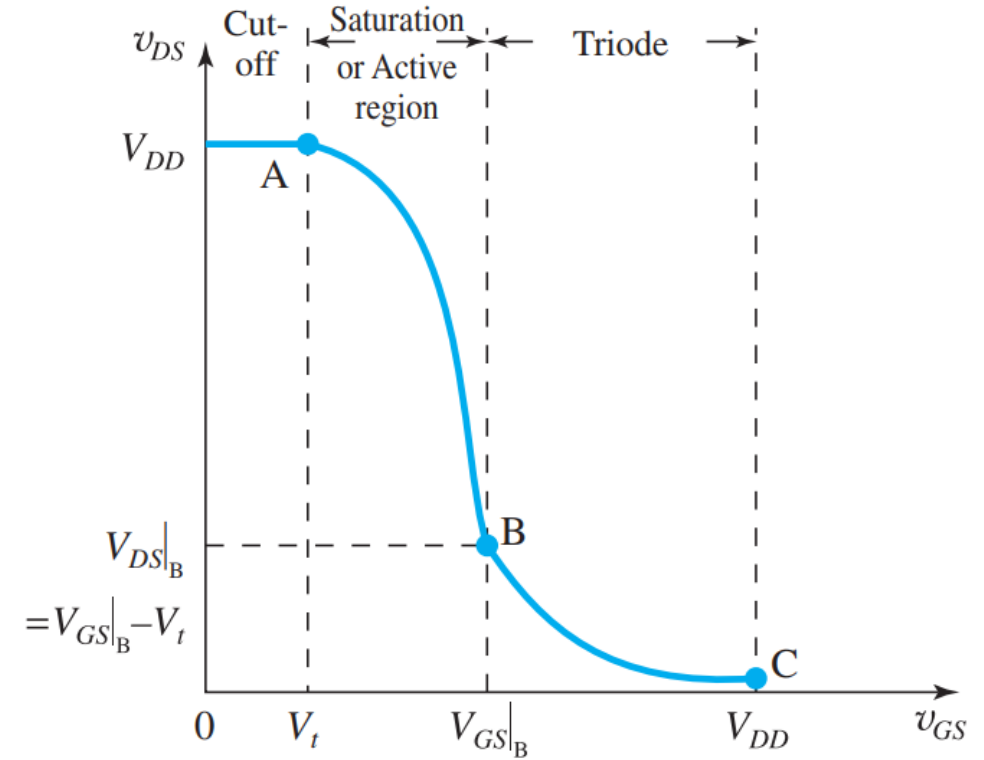
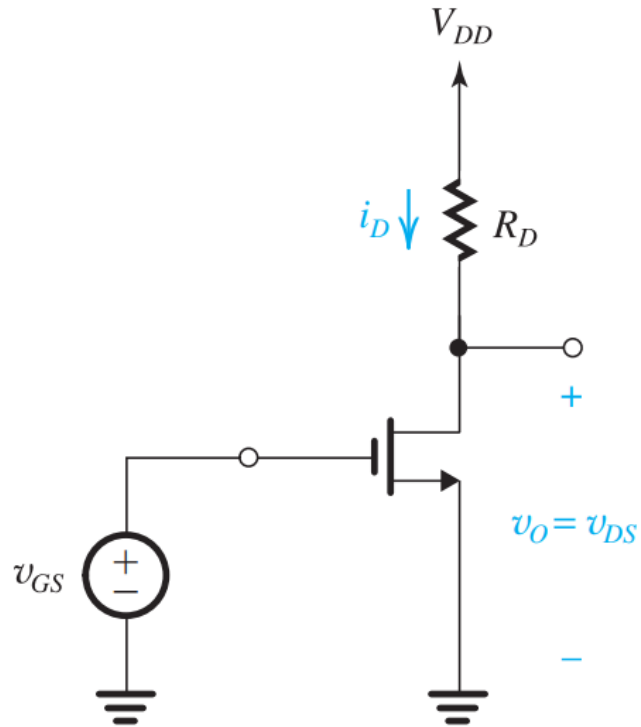
$$V_t = V_{t0} + \gamma \left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f} \right]$$

where V_{t0} is the threshold voltage for $V_{SB} = 0$; ϕ_f is a physical parameter with $(2\phi_f)$ typically 0.6 V; γ is a fabrication-process parameter given by

$$\gamma = \frac{\sqrt{2qN_A\epsilon_s}}{C_{ox}}$$

where q is the magnitude of the electron charge (1.6×10^{-19} C), N_A is the doping concentration of the p -type substrate, and ϵ_s is the permittivity of silicon ($11.7\epsilon_0 = 11.7 \times 8.854 \times 10^{-14} = 1.04 \times 10^{-12}$ F/cm). The parameter γ is known as the **body-effect parameter**.

Amplifiers



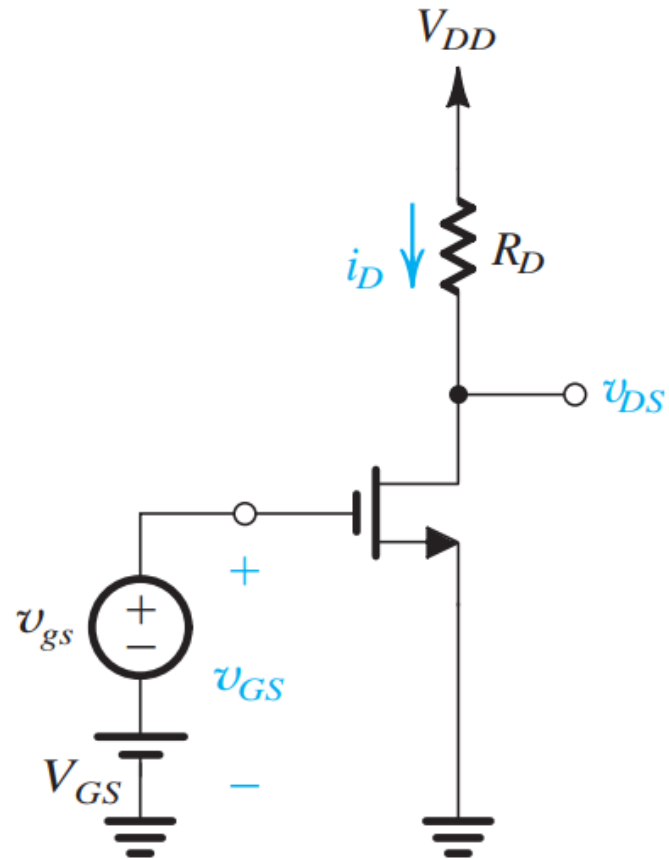
$$v_{DS} = V_{DD} - i_D R_D$$

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2$$

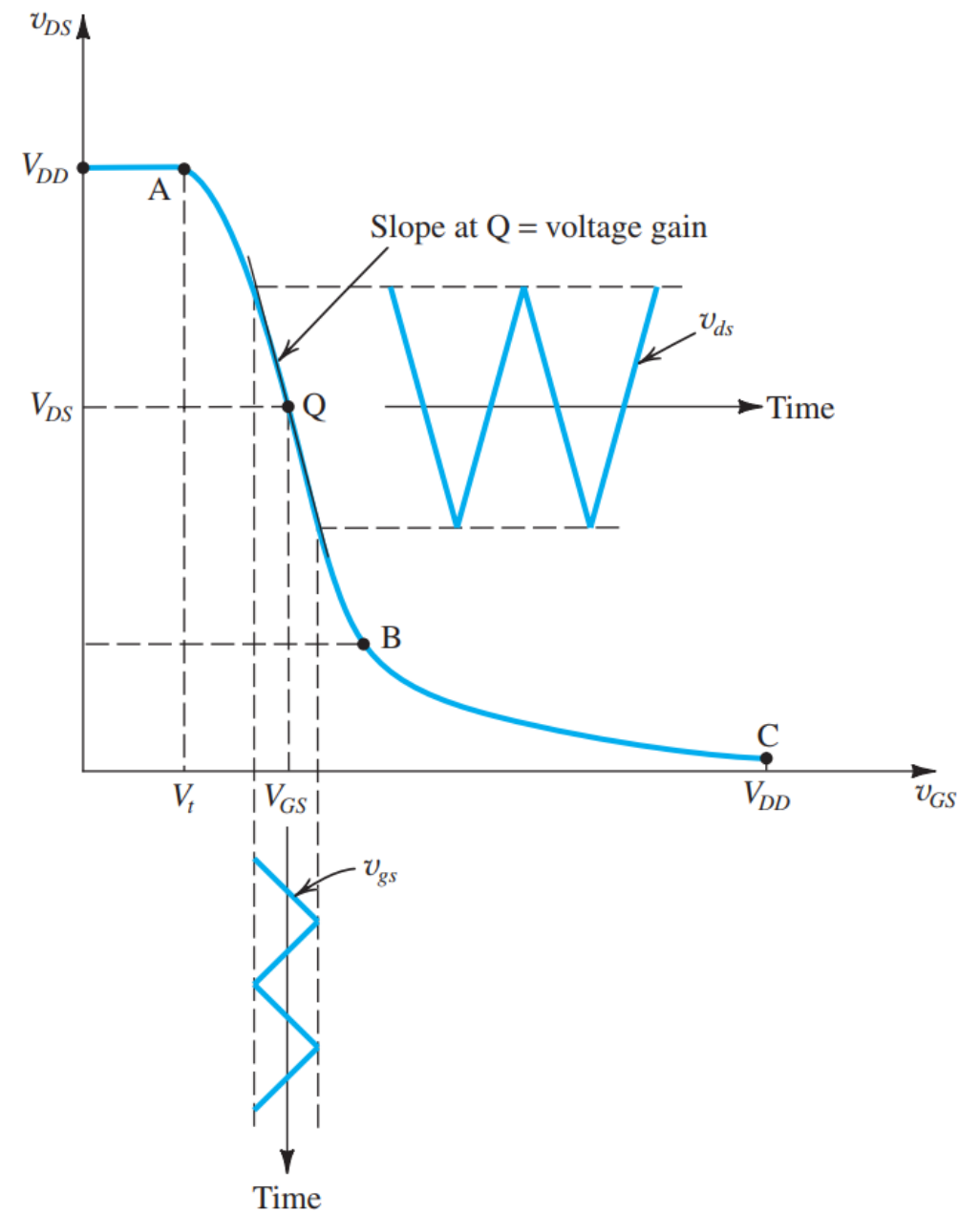
$$V_{DS}|_B = V_{OV}|_B \quad V_{GS}|_B = V_t + \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D}$$

$$V_{OV}|_B \equiv V_{GS}|_B - V_t = \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D}$$

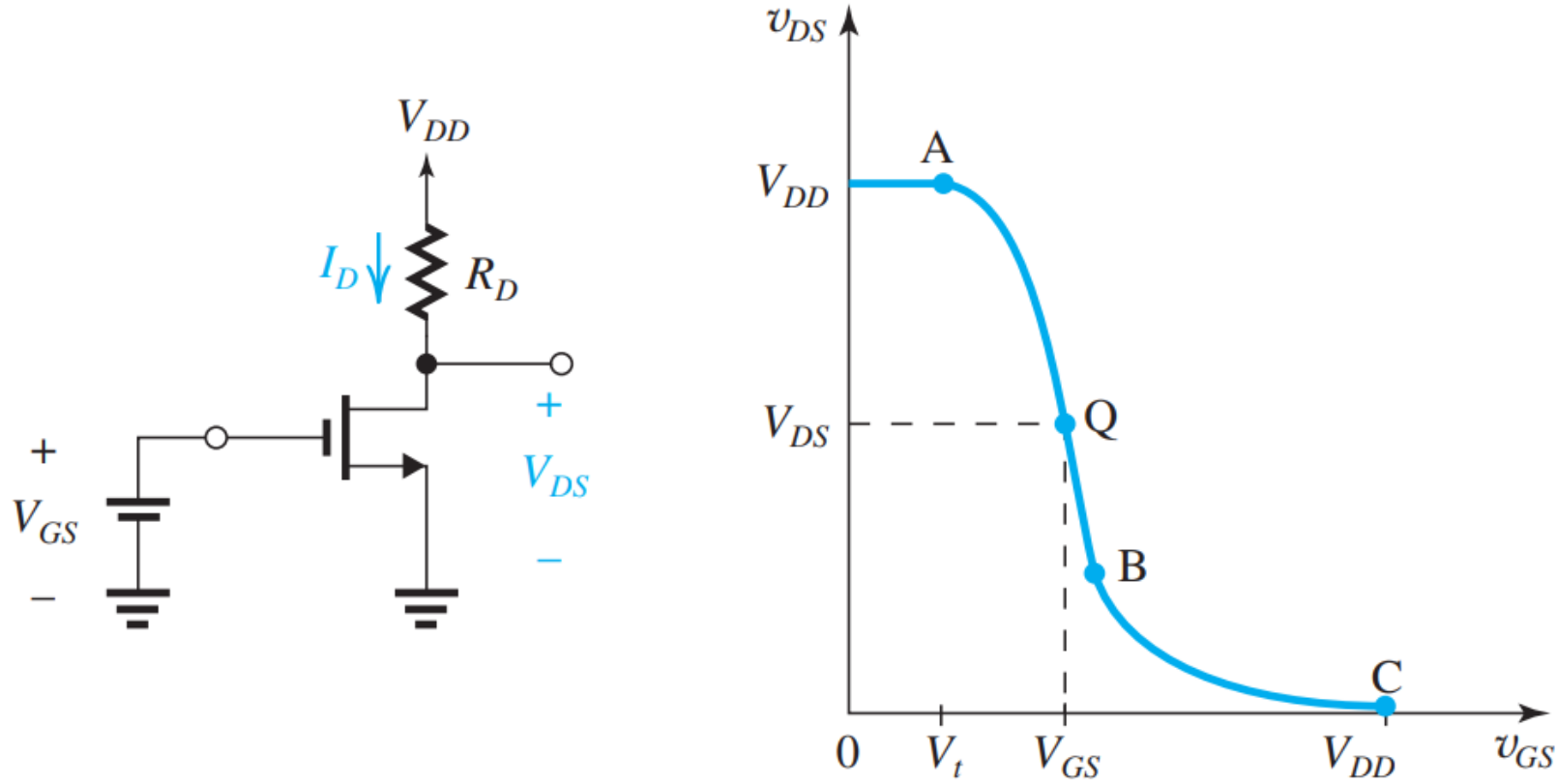
Amplifiers



$$v_{GS}(t) = V_{GS} + v_{gs}(t)$$



Amplifiers



$$V_{DS} = V_{DD} - \frac{1}{2}k_n R_D (V_{GS} - V_t)^2$$

Small Signal Voltage Gain

$$A_v = \left. \frac{dv_{DS}}{dv_{GS}} \right|_{v_{GS}=V_{GS}}$$

$$A_v = -k_n(V_{GS} - V_t)R_D$$

$$A_v = -k_n V_{OV} R_D$$

$$I_D = \frac{1}{2} k_n V_{OV}^2$$

$$A_v = -\frac{I_D R_D}{V_{OV}/2}$$

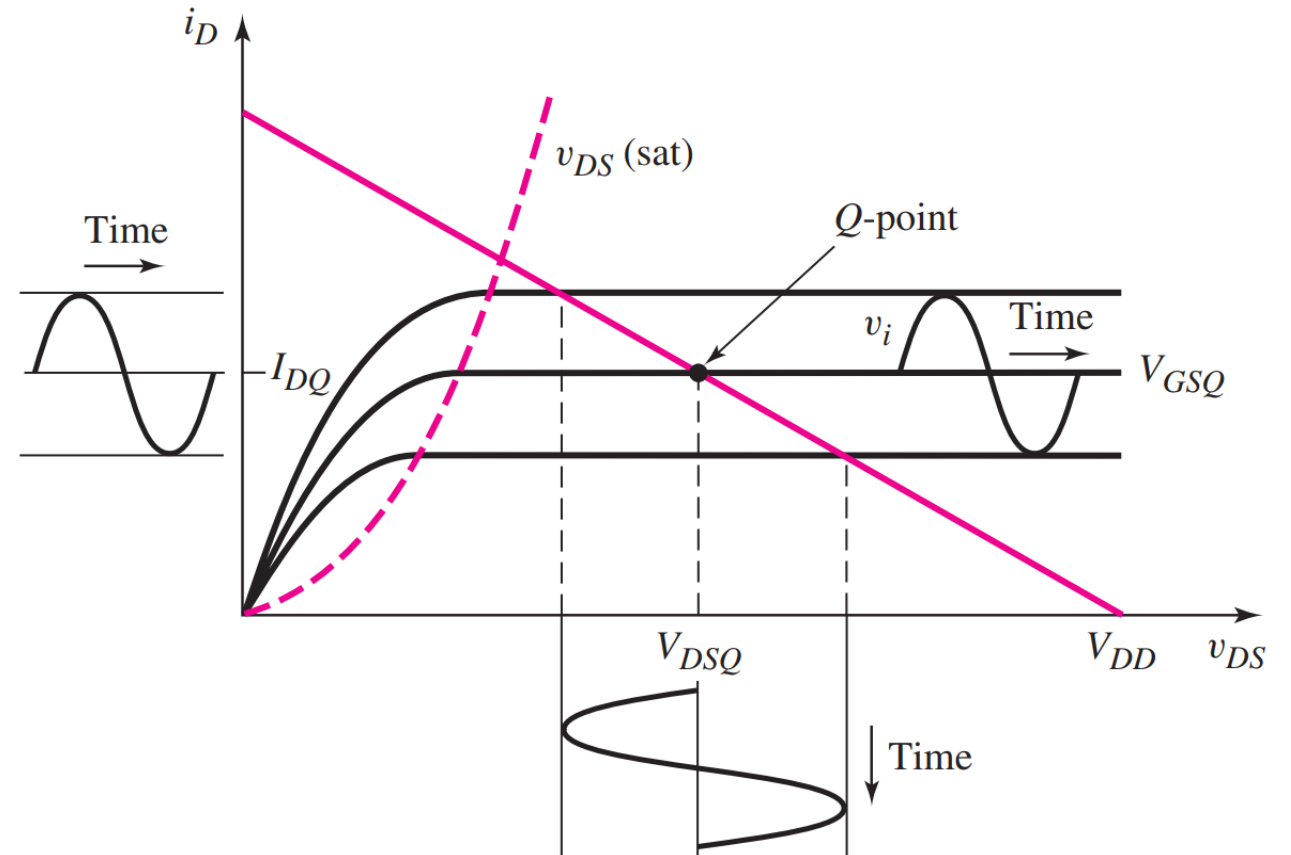
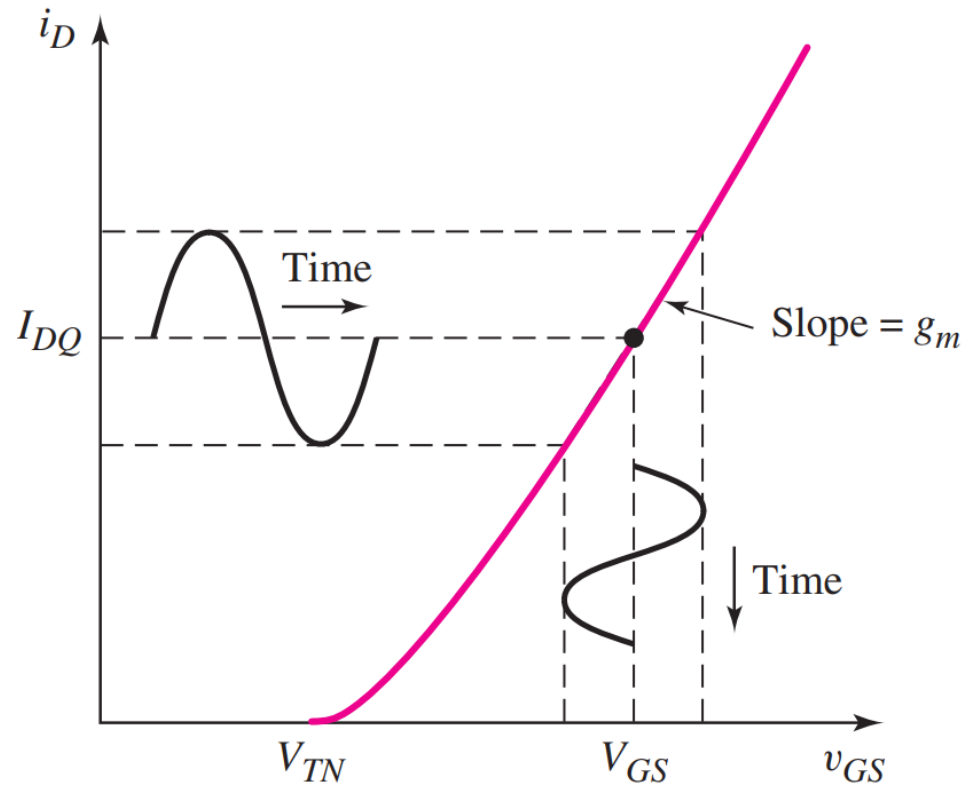
$$A_v = -\frac{V_{DD} - V_{DS}}{V_{OV}/2}$$

$$|A_{v\max}| = \frac{V_{DD} - V_{DS}|_B}{V_{OV}|_B/2}$$

$$|A_{v\max}| = \frac{V_{DD} - V_{OV}|_B}{V_{OV}|_B/2}$$

1. The gain is negative, which signifies that the amplifier is inverting; that is, there is a 180° phase shift between the input and the output. This inversion is obvious in Fig. 7.4(b) and should have been anticipated from Eq. (7.5).
2. The gain is proportional to the load resistance R_D , to the transistor transconductance parameter k_n , and to the overdrive voltage V_{OV} . This all makes intuitive sense.

Graphical Analysis



Small Signal Voltage Gain

$$I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2 = \frac{1}{2}k_n V_{OV}^2$$

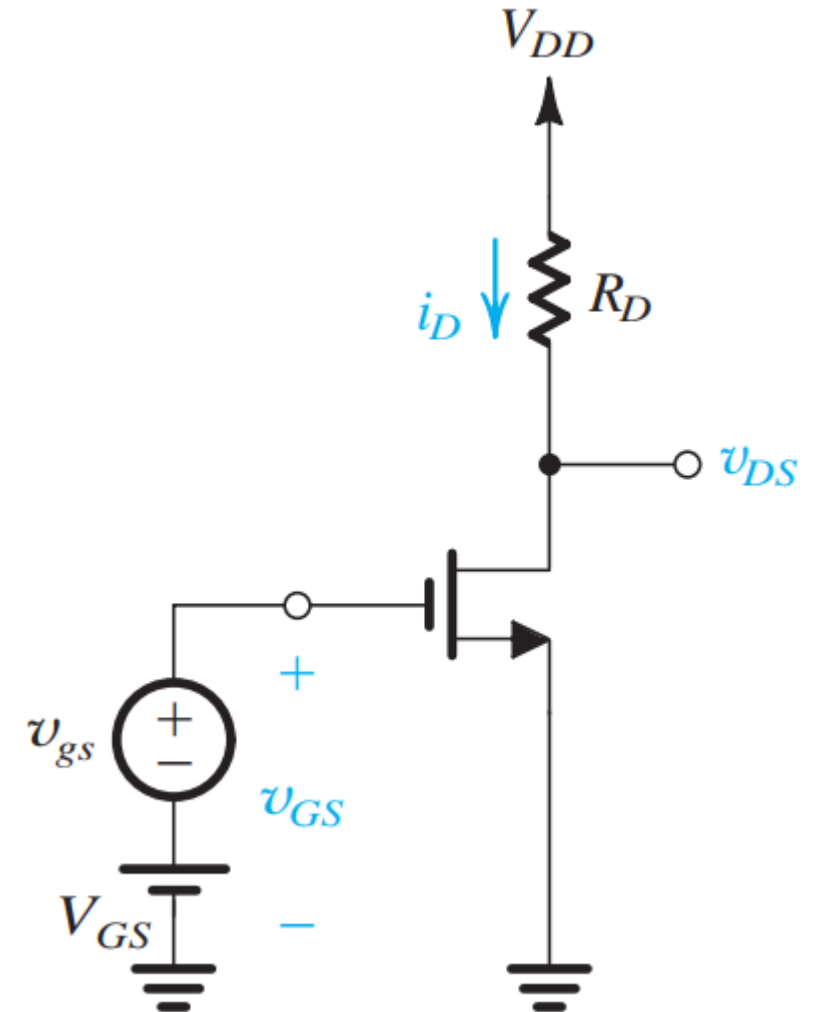
$$V_{DS} = V_{DD} - R_D I_D$$

To ensure saturation-region operation, we must have

$$V_{DS} > V_{OV}$$

$$v_{GS} = V_{GS} + v_{gs}$$

$$\begin{aligned} i_D &= \frac{1}{2}k_n(V_{GS} + v_{gs} - V_t)^2 \\ &= \frac{1}{2}k_n(V_{GS} - V_t)^2 + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \end{aligned}$$



Small Signal Voltage Gain

$$\frac{1}{2}k_n v_{gs}^2 \ll k_n (V_{GS} - V_t) v_{gs}$$

$$v_{gs} \ll 2(V_{GS} - V_t) \quad \text{or,} \quad v_{gs} \ll 2V_{OV}$$

$$i_D \simeq I_D + i_d$$

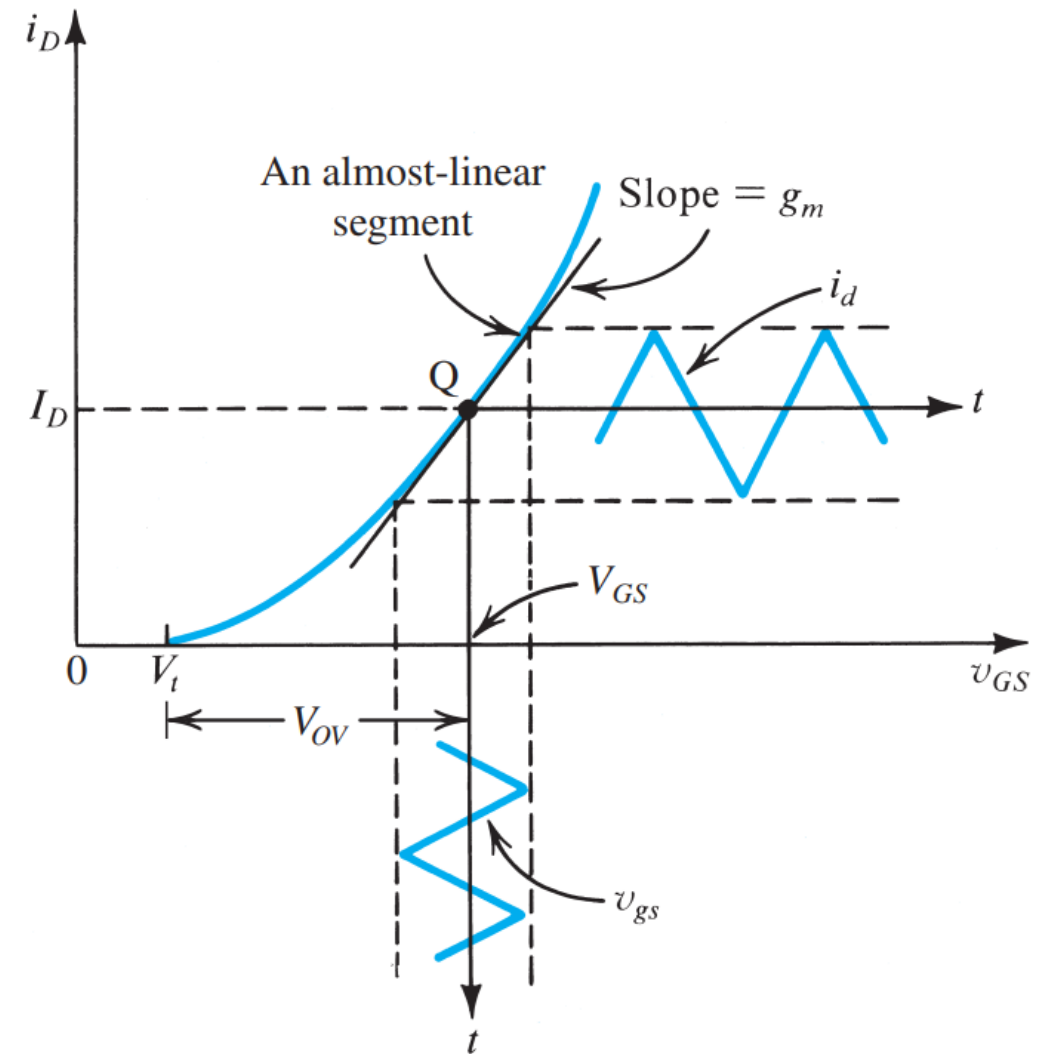
$$i_d = k_n (V_{GS} - V_t) v_{gs}$$

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n (V_{GS} - V_t)$$

$$g_m = k_n V_{OV}$$

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS}=V_{GS}}$$

where g_m is the **transconductance**



Small Signal Voltage Gain

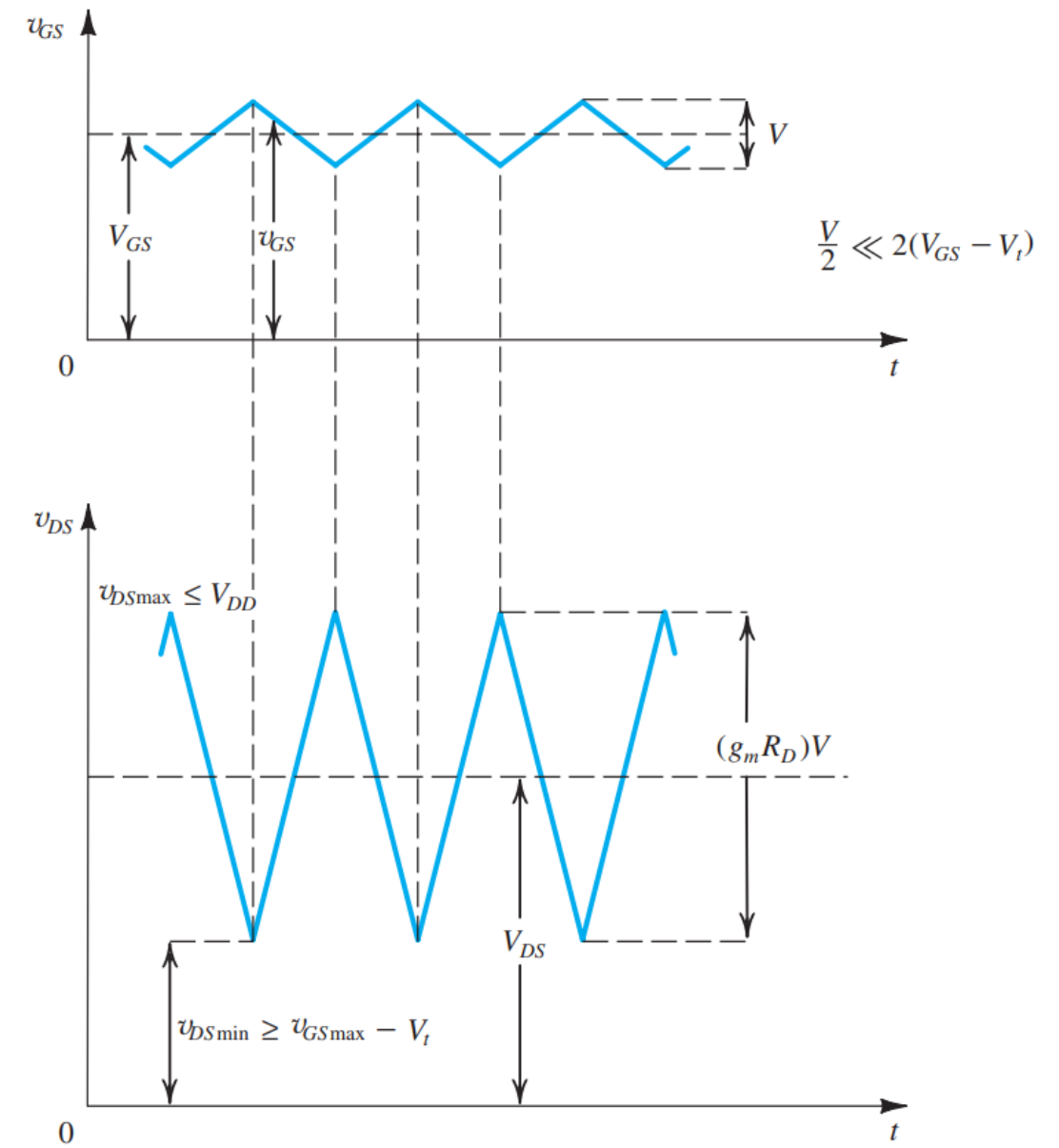
$$v_{DS} = V_{DD} - R_D i_D$$

$$v_{DS} = V_{DD} - R_D (I_D + i_d)$$

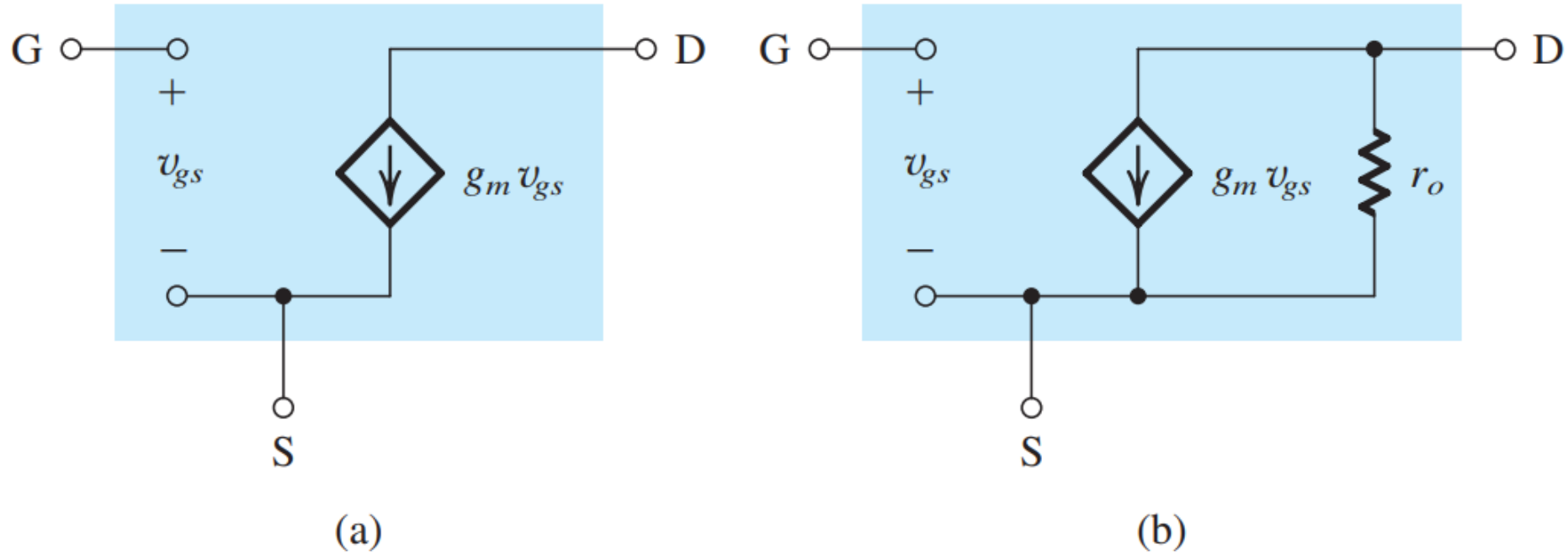
$$v_{DS} = V_{DS} - R_D i_d$$

$$v_{ds} = -i_d R_D = -g_m v_{gs} R_D$$

$$A_v \equiv \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

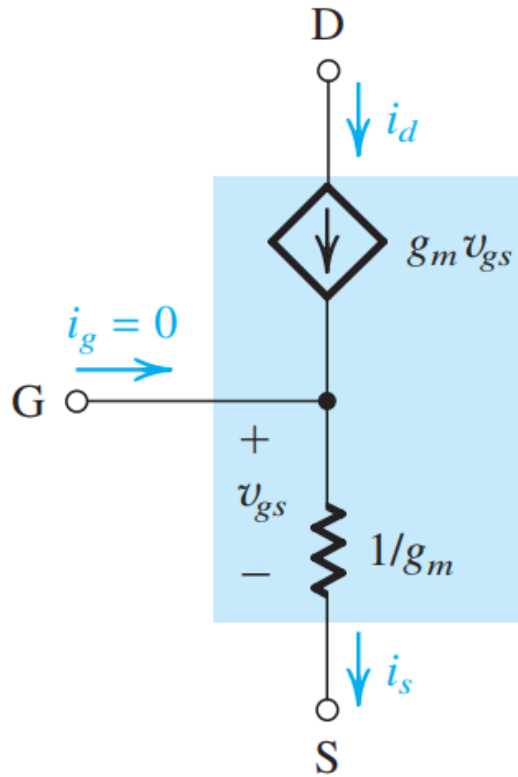


Small Signal Models (Hybrid- π Model)

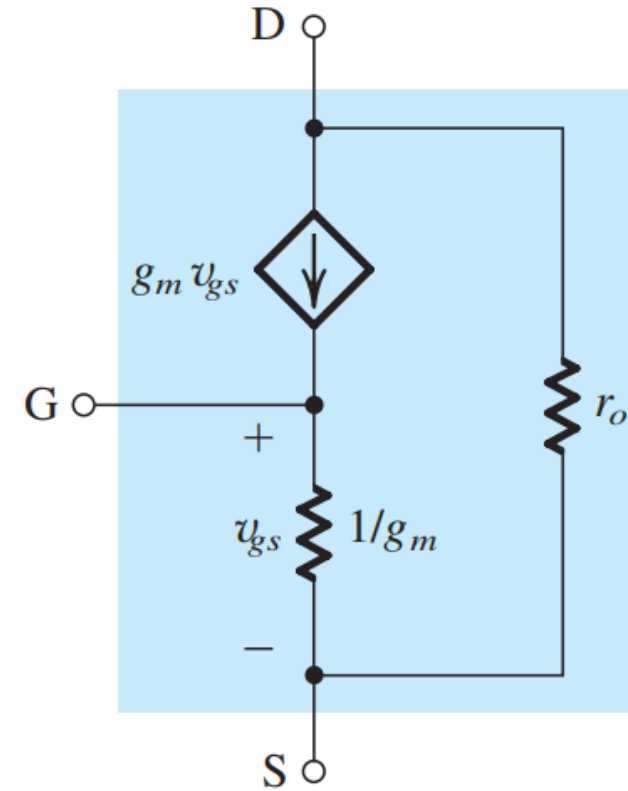


$$r_o = \frac{|V_A|}{I_D}$$

Small Signal Models (T Model)



(a)



(b)

$$r_o = \frac{|V_A|}{I_D}$$

Small Signal Voltage Gain

$$A_v = \frac{v_{ds}}{v_{gs}} = -g_m(R_D \parallel r_o)$$

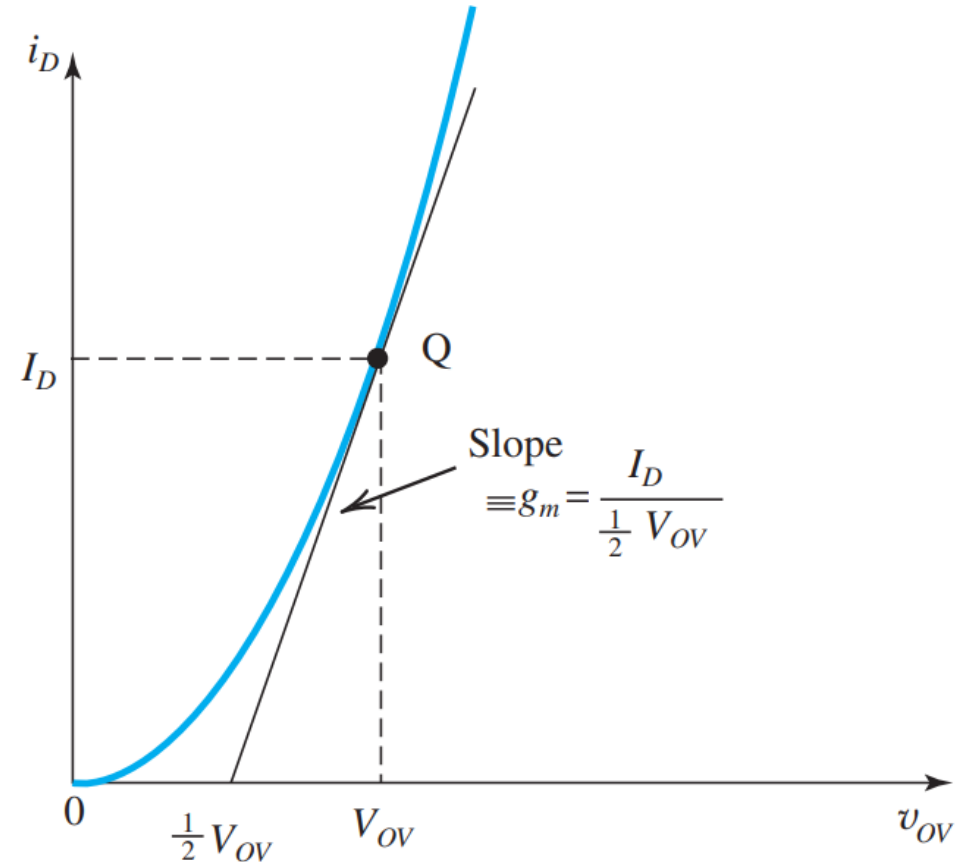
$$g_m = k'_n(W/L)(V_{GS} - V_t) = k'_n(W/L)V_{OV}$$

$$g_m = \sqrt{2k'_n} \sqrt{W/L} \sqrt{I_D}$$

$$g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}}$$

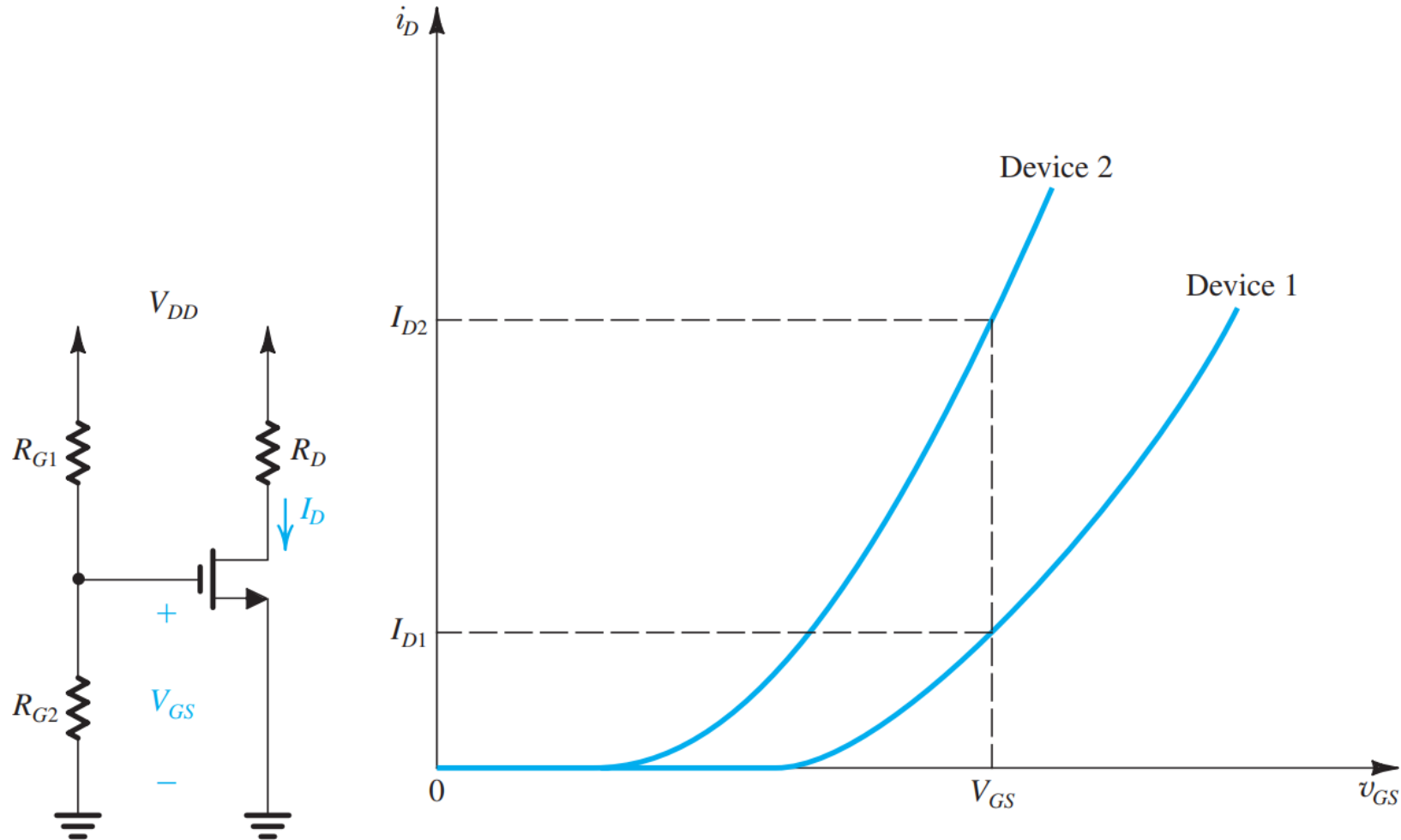
This expression shows two things:

1. For a given MOSFET, g_m is proportional to the square root of the dc bias current.
2. At a given bias current, g_m is proportional to $\sqrt{W/L}$.



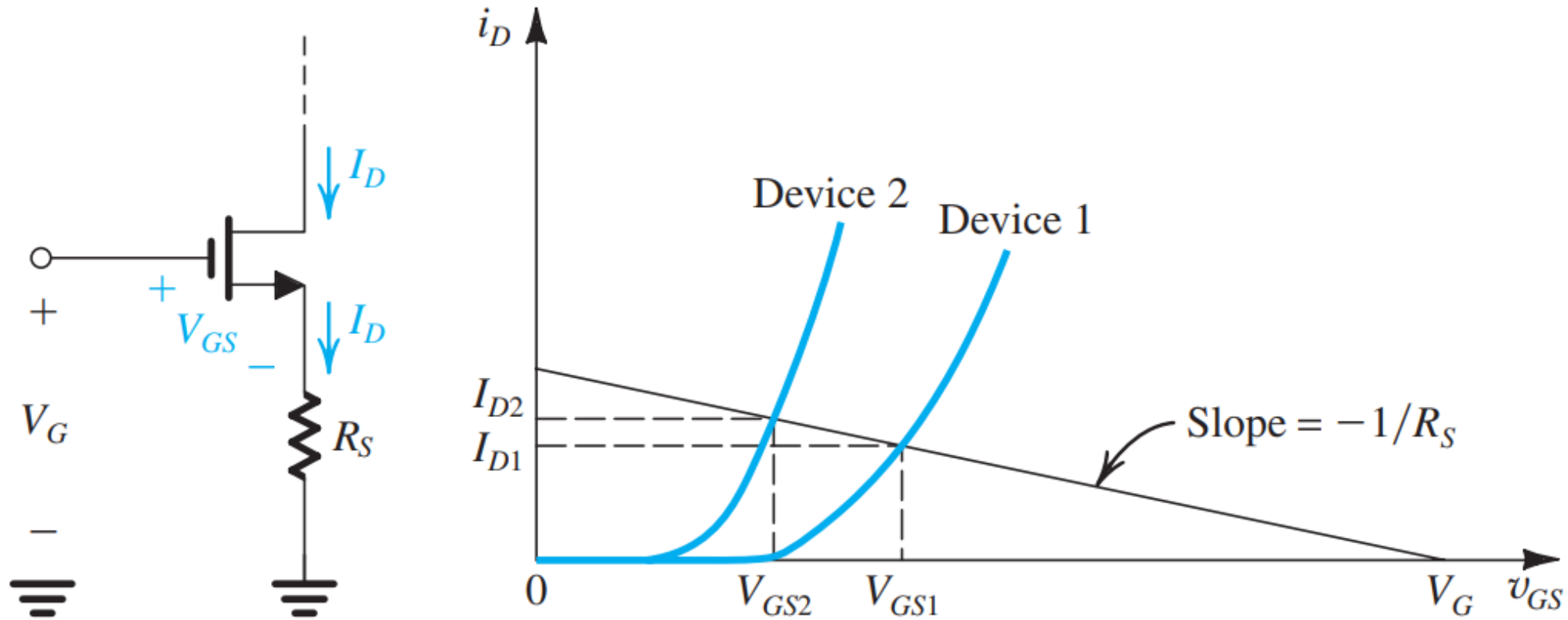
Biasing Circuit

Biasing by Fixing V_{GS}



Biasing Circuit

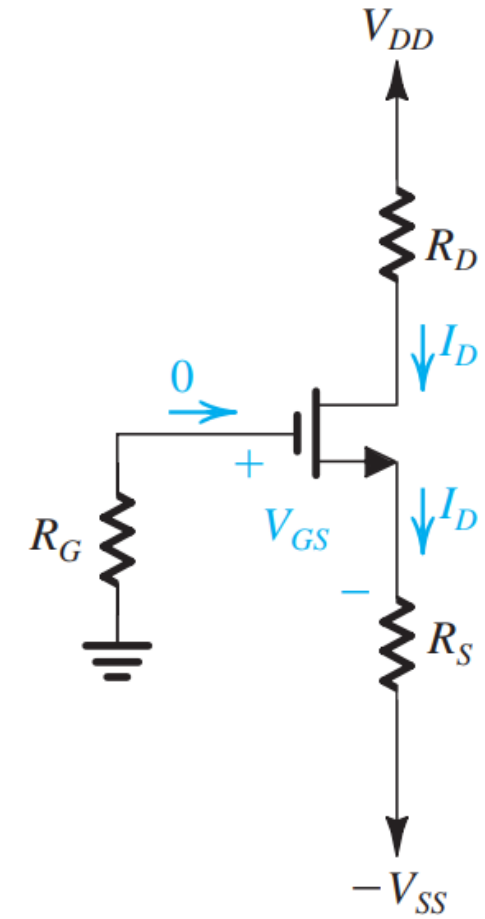
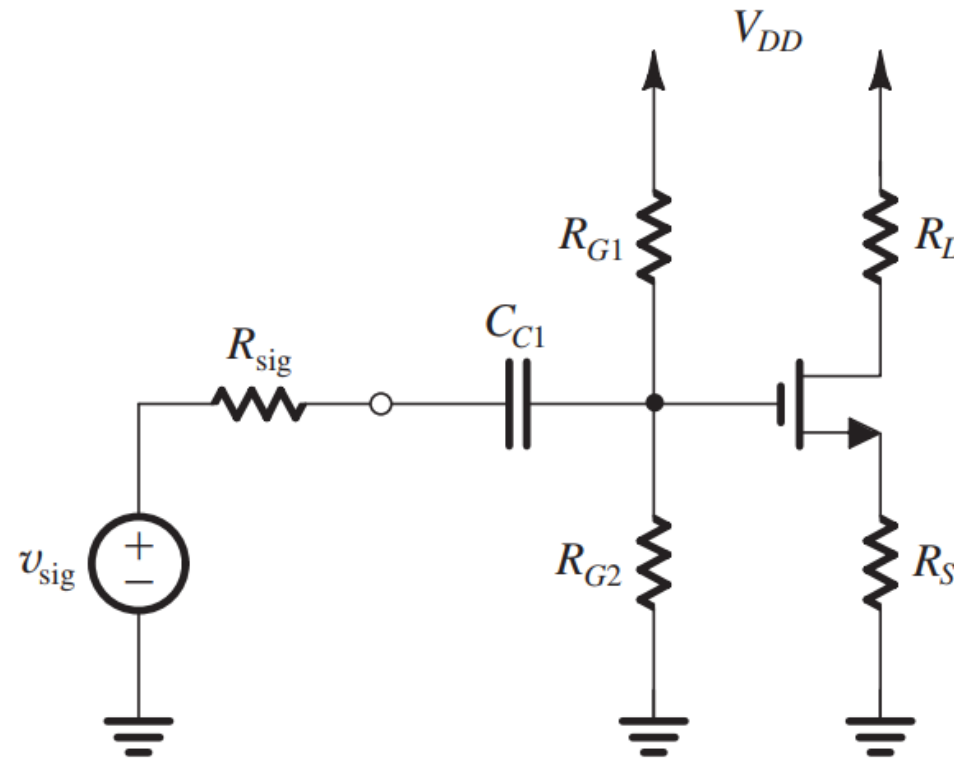
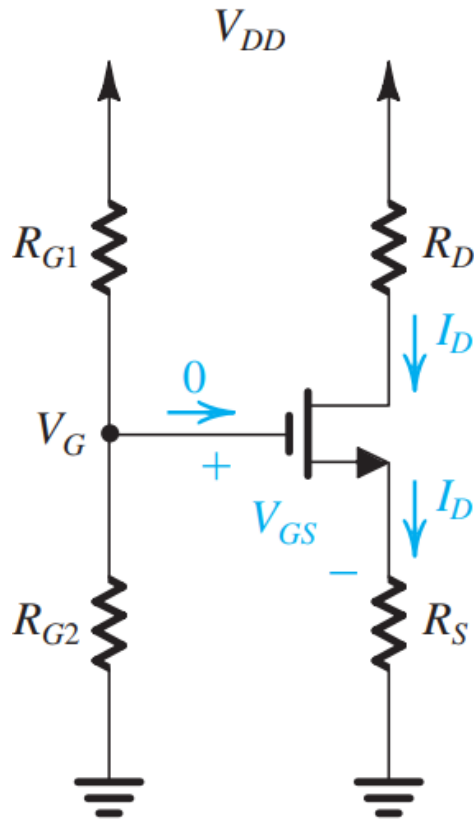
Biasing by Fixing V_G and Connecting a Resistance in the Source



$$V_G = V_{GS} + R_S I_D$$

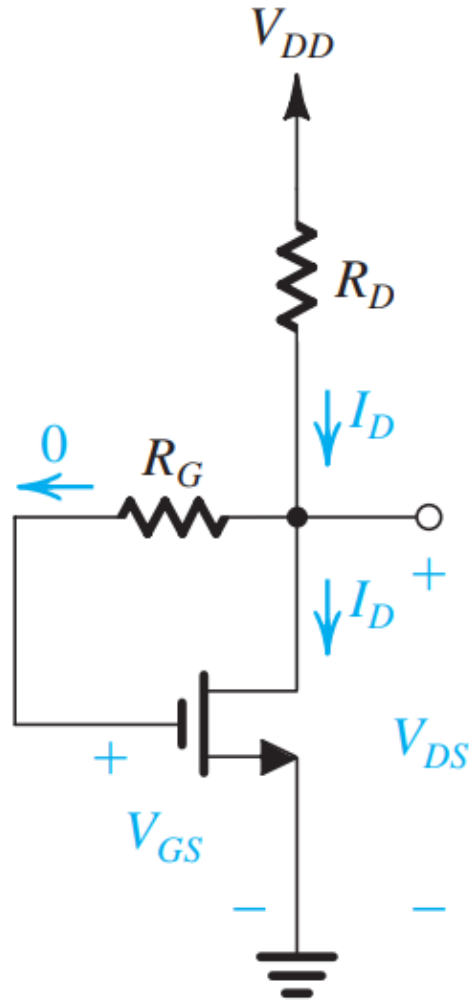
Biassing Circuit

Biassing by Fixing V_G and Connecting a Resistance in the Source



Biasing Circuit

Biasing Using a Drain-to-Gate Feedback Resistor



$$V_{GS} = V_{DS} = V_{DD} - R_D I_D$$

$$V_{DD} = V_{GS} + R_D I_D$$

Example

It is required to design the circuit of Fig. 7.48(c) to establish a dc drain current $I_D = 0.5$ mA. The MOSFET is specified to have $V_t = 1$ V and $k'_n W/L = 1$ mA/V². For simplicity, neglect the channel-length modulation effect (i.e., assume $\lambda = 0$). Use a power-supply $V_{DD} = 15$ V. Calculate the percentage change in the value of I_D obtained when the MOSFET is replaced with another unit having the same $k'_n W/L$ but $V_t = 1.5$ V.

Solution

As a rule of thumb for designing this classical biasing circuit, we choose R_D and R_S to provide one-third of the power-supply voltage V_{DD} as a drop across each of R_D , the transistor (i.e., V_{DS}), and R_S . For $V_{DD} = 15$ V, this choice makes $V_D = +10$ V and $V_S = +5$ V. Now, since I_D is required to be 0.5 mA, we can find the values of R_D and R_S as follows:

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{15 - 10}{0.5} = 10 \text{ k}\Omega$$

$$R_S = \frac{V_S}{I_D} = \frac{5}{0.5} = 10 \text{ k}\Omega$$

$$I_D = \frac{1}{2} k'_n (W/L) V_{OV}^2$$

$$0.5 = \frac{1}{2} \times 1 \times V_{OV}^2$$

$$V_{GS} = V_t + V_{OV} = 1 + 1 = 2 \text{ V}$$

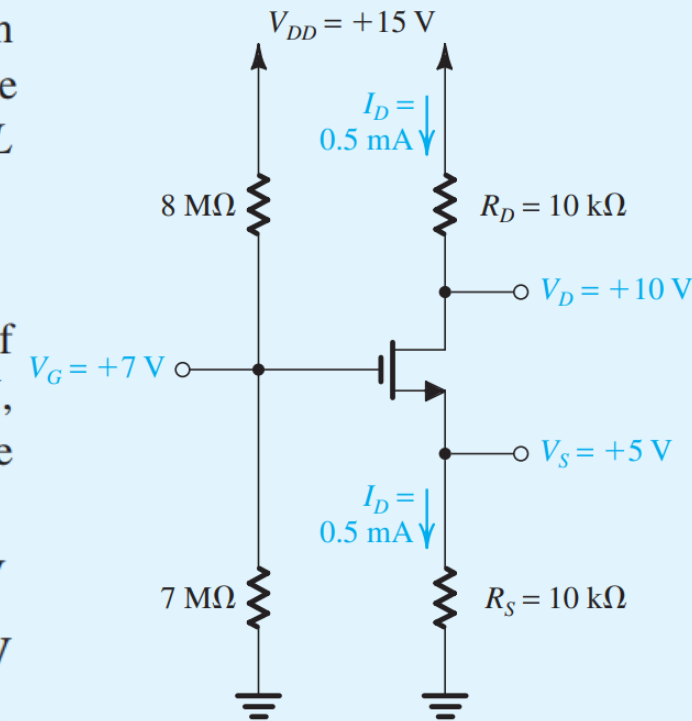
$$V_G = V_S + V_{GS} = 5 + 2 = 7 \text{ V}$$

$$R_{G1} = 8 \text{ M}\Omega \text{ and } R_{G2} = 7 \text{ M}\Omega.$$

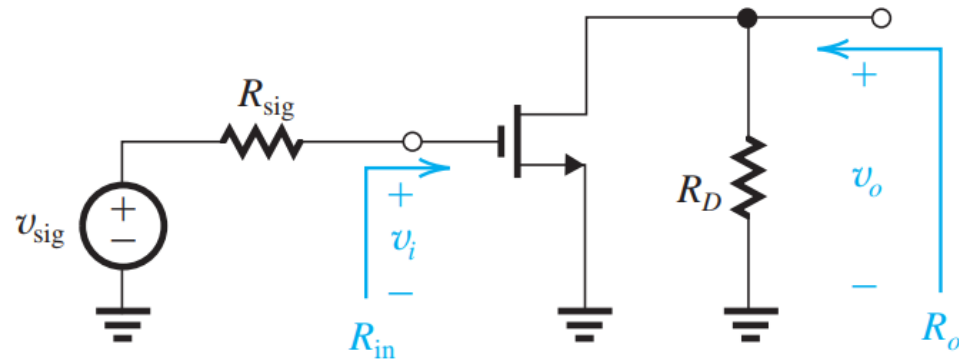
$$I_D = 0.455 \text{ mA}$$

$$\Delta I_D = 0.455 - 0.5 = -0.045 \text{ mA}$$

$$\text{which is } \frac{-0.045}{0.5} \times 100 = -9\% \text{ change}$$



Common-Source Amplifier (Bias Omitted)



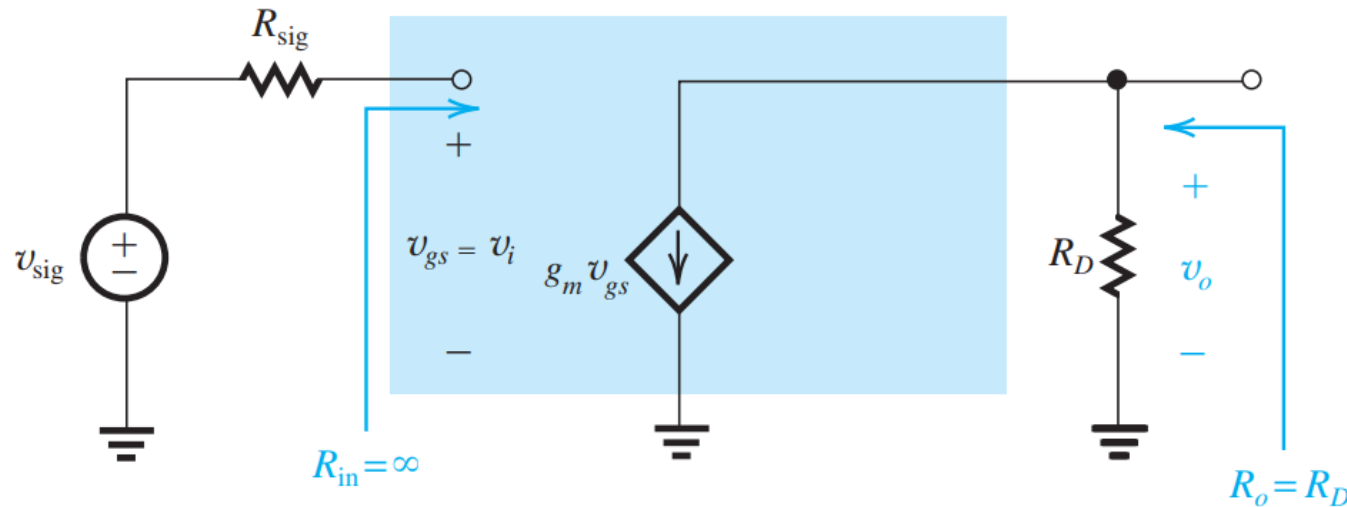
(a)

$$R_{in} = \infty$$

$$v_i = v_{sig}$$

$$v_{gs} = v_i$$

$$v_o = -g_m v_{gs} R_D$$

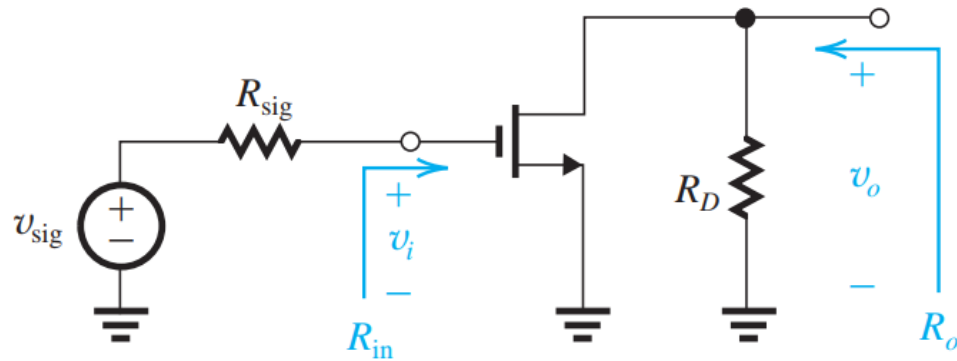


(b)

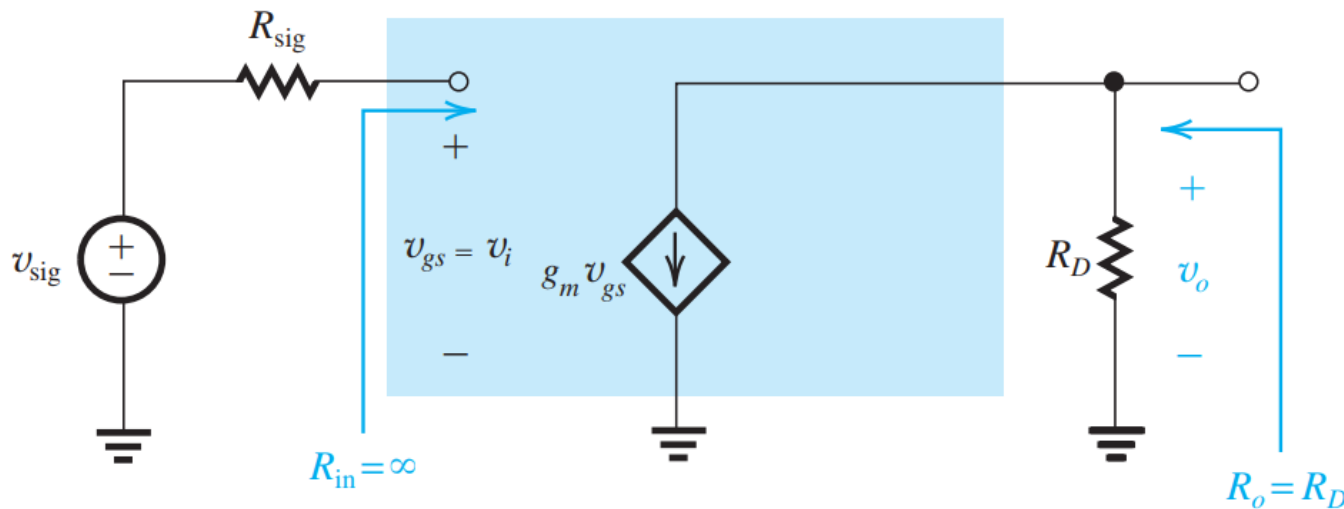
$$A_{vo} \equiv \frac{v_o}{v_i} = -g_m R_D$$

$$R_o = R_D$$

Common-Source Amplifier (Bias Omitted)



(a)



(b)

With a load resistance R_L connected across R_D ,

$$A_v = A_{vo} \frac{R_L}{R_L + R_o}$$

$$A_v = -g_m (R_D \parallel R_L)$$

$$R_o = R_D \parallel R_L$$

$$G_v \equiv \frac{v_o}{v_{sig}} = -g_m (R_D \parallel R_L)$$

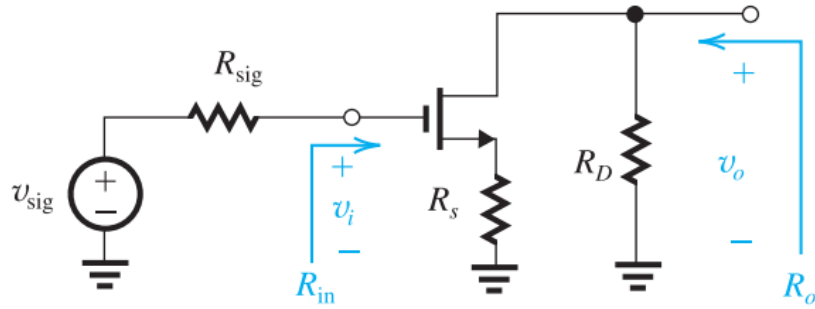
Common-Source Amplifier

A CS amplifier utilizes a MOSFET biased at $I_D = 0.25$ mA with $V_{ov} = 0.25$ V and $R_D = 20$ k Ω . The amplifier is fed with a signal source having $R_{sig} = 100$ k Ω , and a 20-k Ω load is connected to the output. Find R_{in} , A_{vo} , R_o , A_v , and G_v . If, to maintain reasonable linearity, the peak of the input sine-wave signal is limited to 10% of $2V_{ov}$, what is the peak of the sine-wave voltage at the output?

Ans. ∞ ; -40 V/V; 20 k Ω ; -20 V/V; -20 V/V; 1 V



The CS Amplifier with Source Resistance (Bias Omitted)



(a)

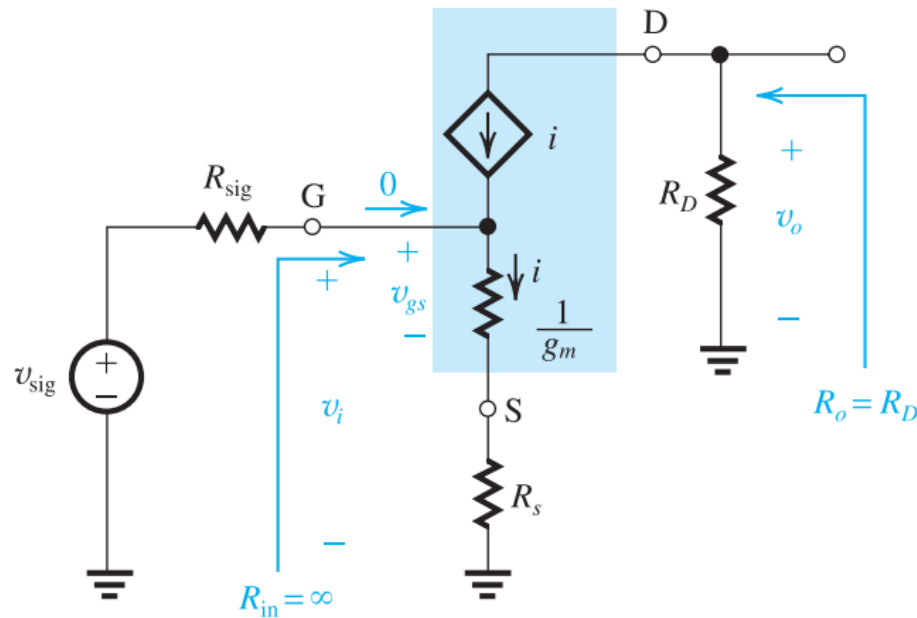
$$v_{gs} = v_i \frac{1/g_m}{1/g_m + R_s} = \frac{v_i}{1 + g_m R_s}$$

$$v_o = -i R_D$$

$$i = \frac{v_i}{1/g_m + R_s} = \left(\frac{g_m}{1 + g_m R_s} \right) v_i$$

$$A_{vo} \equiv \frac{v_o}{v_i} = - \frac{g_m R_D}{1 + g_m R_s}$$

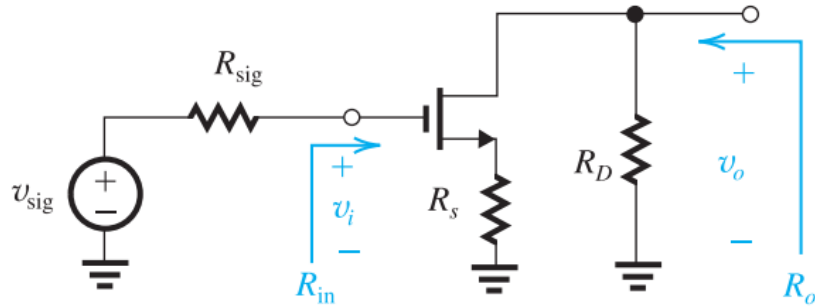
$$A_{vo} = - \frac{R_D}{1/g_m + R_s}$$



(b)

Voltage gain from gate to drain = $-\frac{\text{Total resistance in drain}}{\text{Total resistance in source}}$

The CS Amplifier with Source Resistance (Bias Omitted)



(a)

$$R_{in} = \infty$$

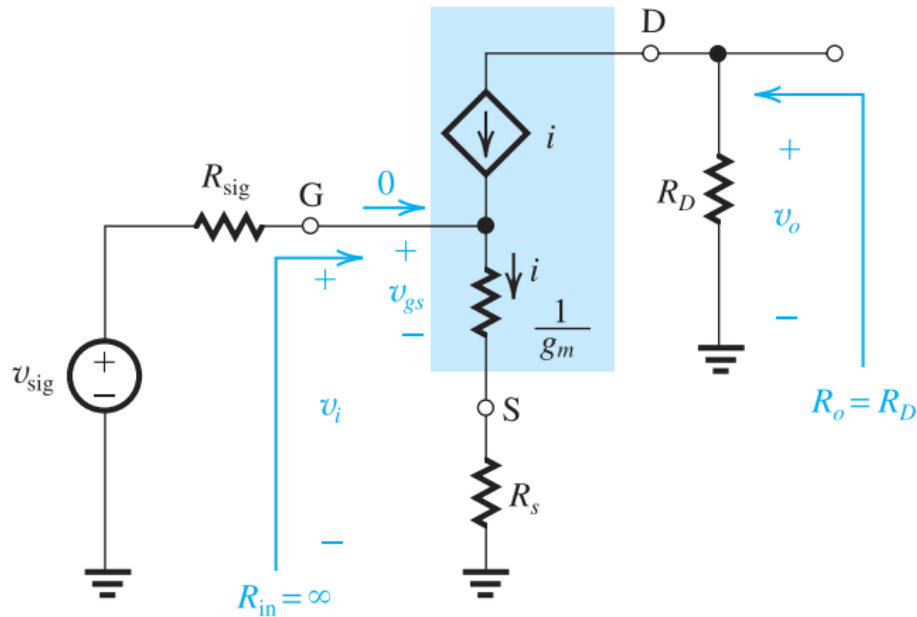
$$R_o = R_D$$

With a load resistance R_L connected across R_D ,

$$A_v = -\frac{g_m(R_D \parallel R_L)}{1 + g_m R_s}$$

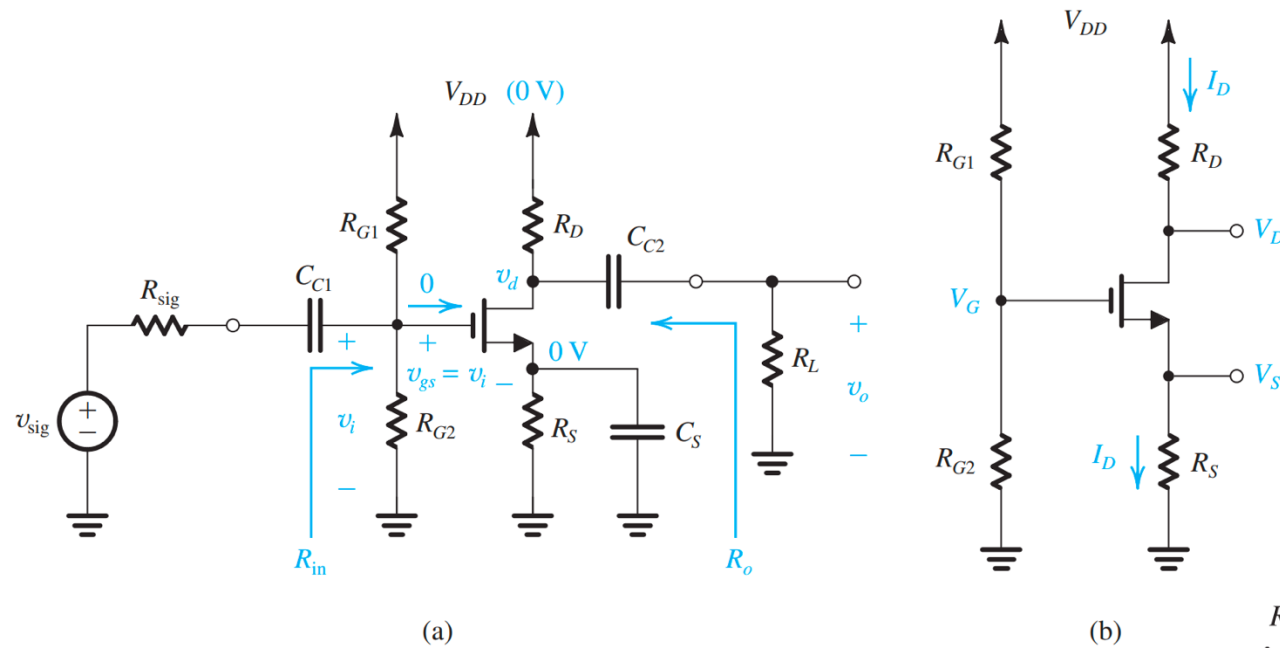
$$A_v = -\frac{R_D \parallel R_L}{1/g_m + R_s}$$

Because R_{in} is infinite, $v_i = v_{sig}$ and the overall voltage gain G_v is equal to A_v



(b)

The CS Amplifier with Source Resistance (Including Bias)



$$A_{vo} = \frac{v_o \text{ (without } R_L)}{v_{gs}} = g_m (R_D \parallel r_o)$$

$$A_v = \frac{v_o \text{ (with } R_L)}{v_{gs}} = g_m (R_D \parallel r_o \parallel R_L)$$

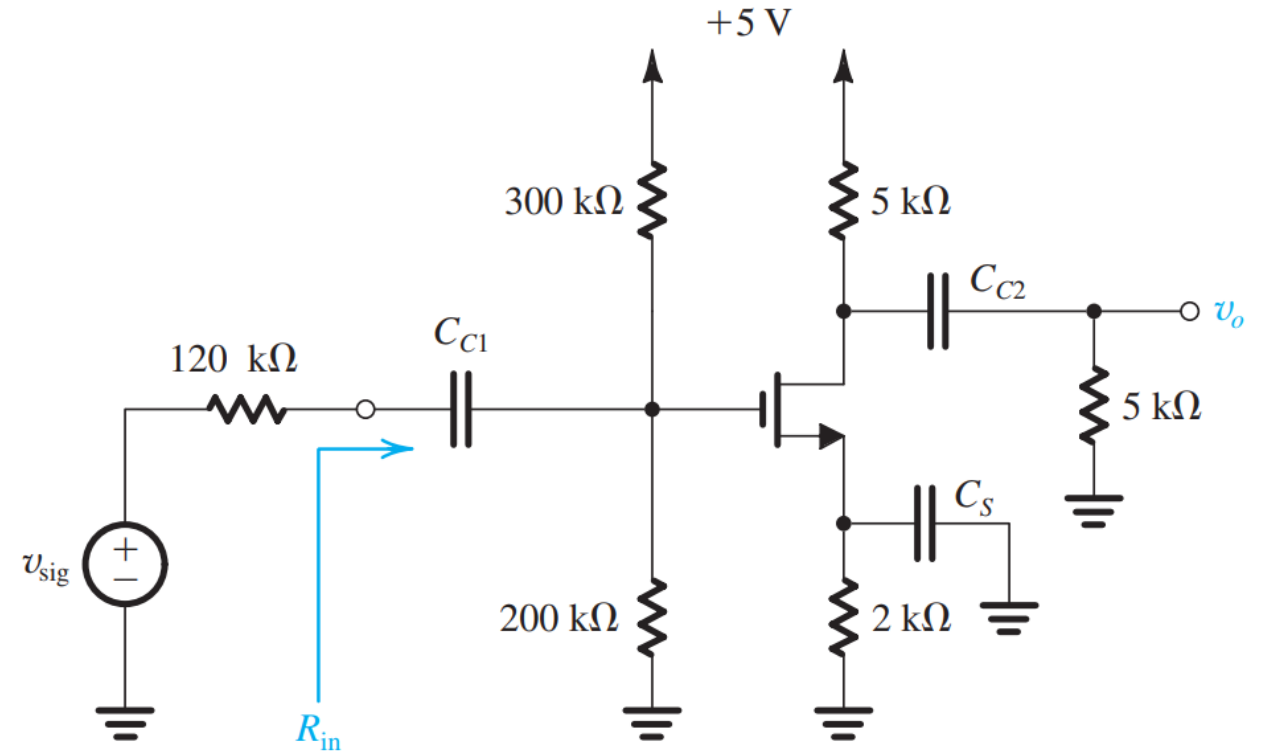
$$G_v = -\frac{R_{in}}{R_{in} + R_{sig}} g_m (R_D \parallel R_L \parallel r_o)$$

Figure 7.55 (a) A common-source amplifier using the classical biasing arrangement of Fig. 7.48(c). (b) Circuit for determining the bias point. (c) Equivalent circuit and analysis.

The CS Amplifier Example

The NMOS transistor in the CS amplifier shown in Fig. P7.118 has $V_t = 0.7$ V and $V_A = 50$ V.

- (a) Neglecting the Early effect, verify that the MOSFET is operating in saturation with $I_D = 0.5$ mA and $V_{OV} = 0.3$ V. What must the MOSFET's k_n be? What is the dc voltage at the drain?
- (b) Find R_{in} and G_v .



The CG Amplifier (Bias Omitted)

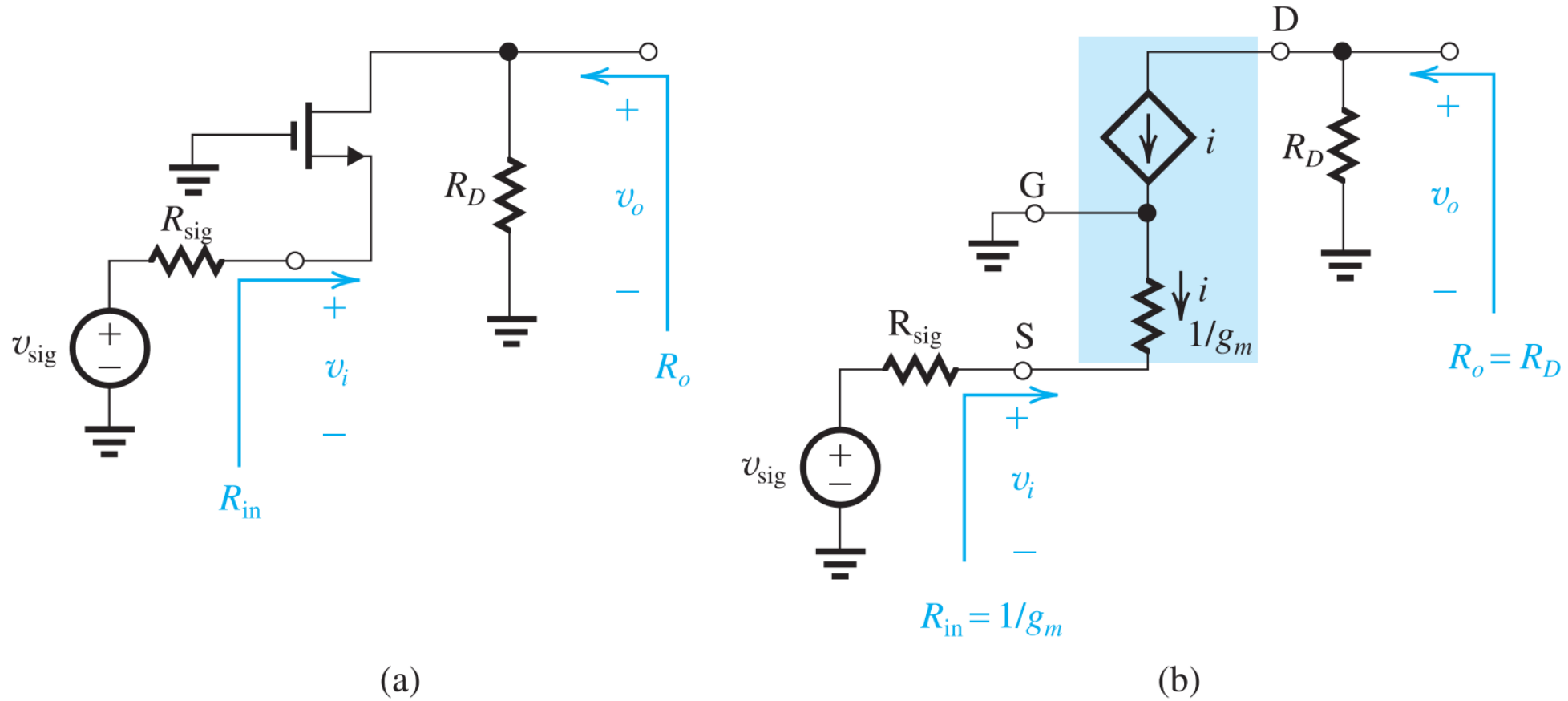
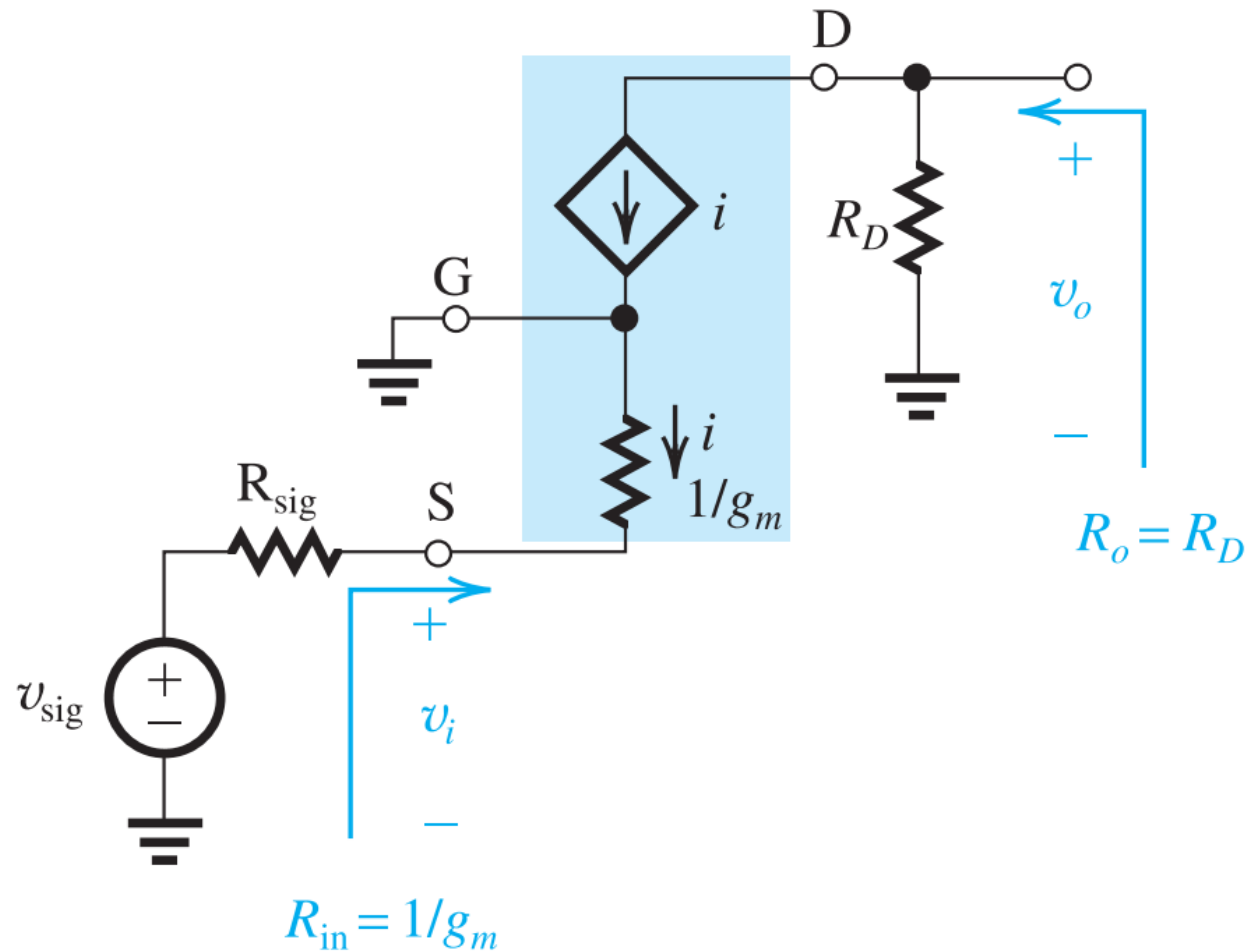


Figure 7.39 (a) Common-gate (CG) amplifier with bias arrangement omitted. (b) Equivalent circuit of the CG amplifier with the MOSFET replaced with its T model.

The CG Amplifier (Bias Omitted)



$$v_o = -iR_D$$

$$i = -\frac{v_i}{1/g_m}$$

$$A_{vo} \equiv \frac{v_o}{v_i} = g_m R_D$$

$$\frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = \frac{1/g_m}{1/g_m + R_{sig}}$$

$$G_v = \frac{1/g_m}{R_{sig} + 1/g_m} [g_m (R_D \parallel R_L)]$$

The Source Follower (CD) Amplifier (Bias Omitted)

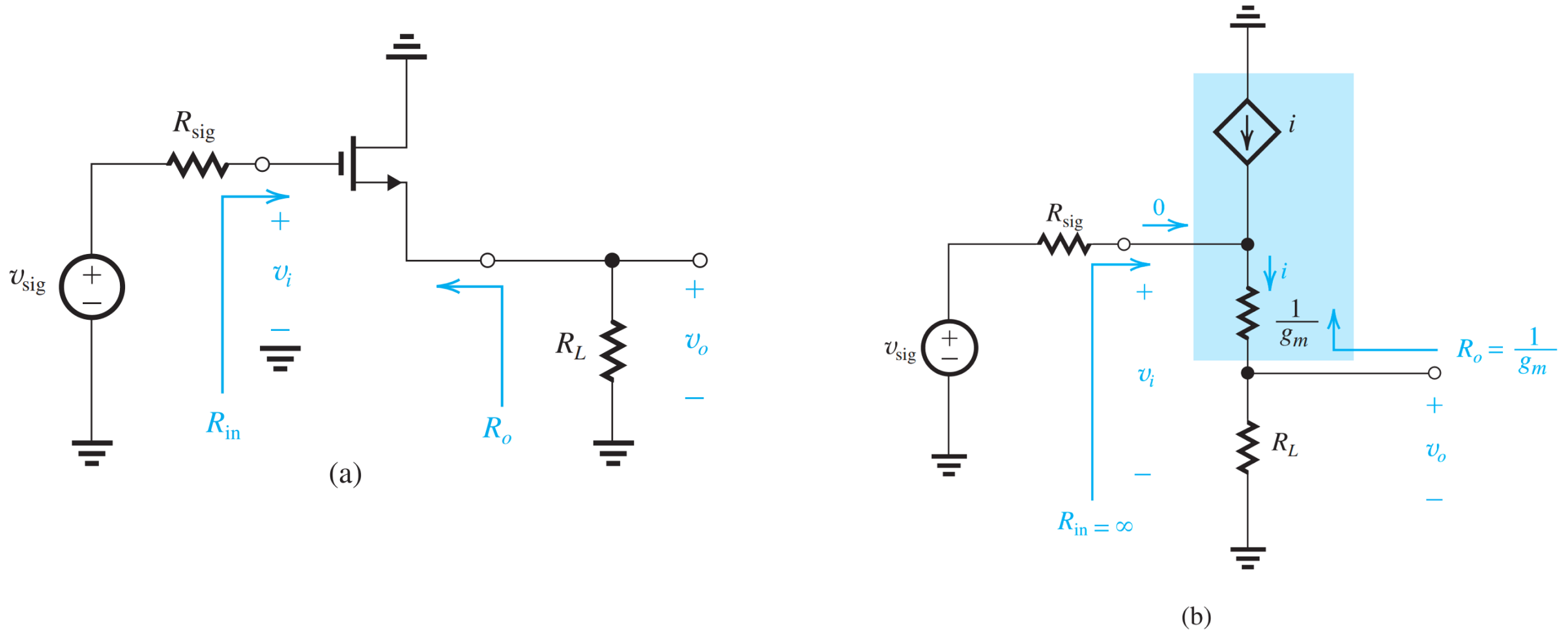
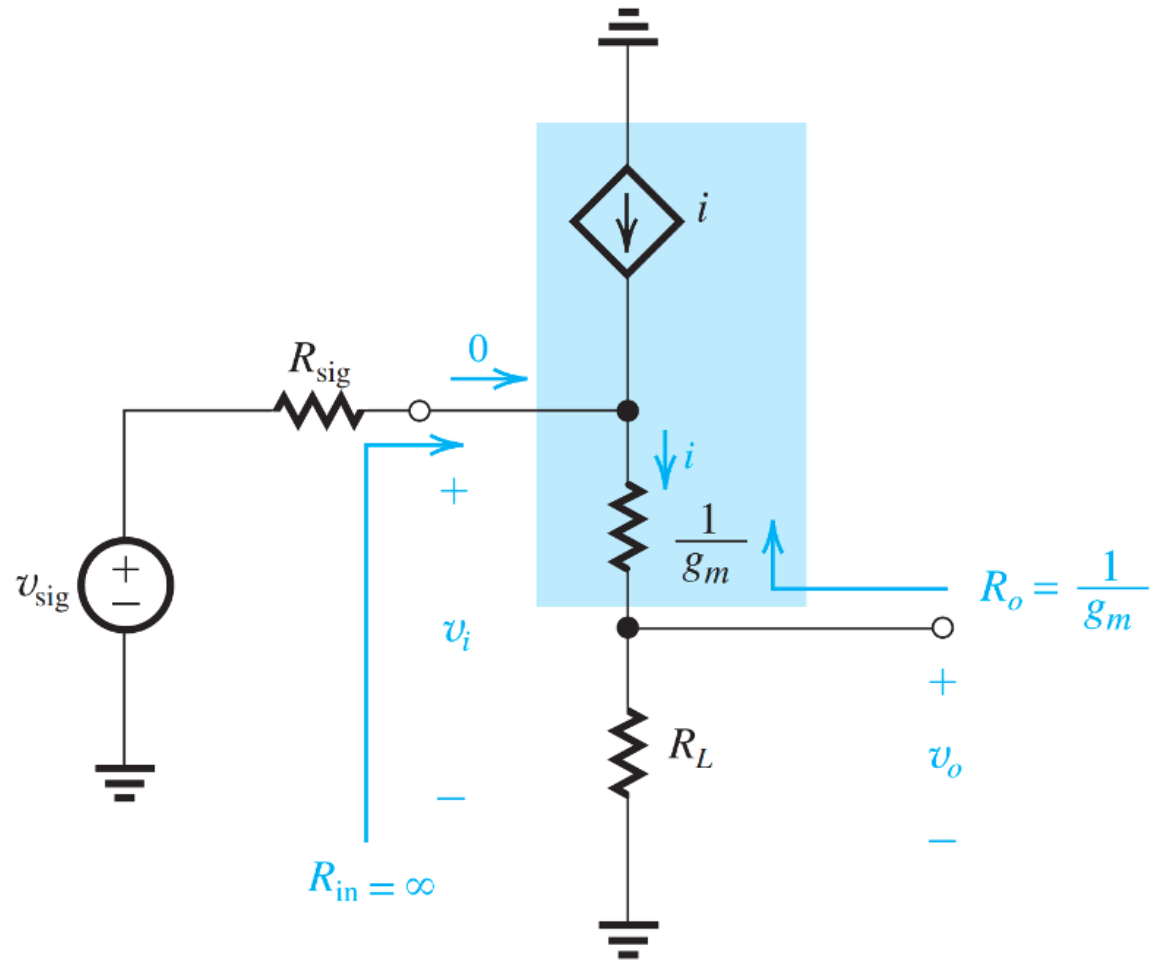


Figure 7.42 (a) Common-drain amplifier or source follower with the bias circuit omitted. (b) Equivalent circuit of the source follower obtained by replacing the MOSFET with its T model.

The Source Follower (CD) Amplifier (Bias Omitted)



$$A_v \equiv \frac{v_o}{v_i} = \frac{R_L}{R_L + 1/g_m}$$

$$A_{vo} = 1$$

$$G_v = A_v = \frac{R_L}{R_L + 1/g_m}$$